Kimodo: Scaling Controllable Human Motion Generation

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https://research.nvidia.com/labs/sil/projects/kimodo

Abstract

High-quality human motion data is becoming increasingly important for applications in robotics, simulation, and entertainment. Recent generative models offer a potential data source, enabling human motion synthesis through intuitive inputs like text prompts or kinematic constraints on poses. However, the small scale of public mocap datasets has limited the motion quality, control accuracy, and generalization of these models. In this work, we introduce Kimodo, an expressive and controllable kinematic motion diffusion model trained on 700 hours of optical motion capture data. Our model generates high-quality motions while being easily controlled through text and a comprehensive suite of kinematic constraints including full-body keyframes, sparse joint positions/rotations, 2D waypoints, and dense 2D paths. This is enabled through a carefully designed motion representation and two-stage denoiser architecture that decomposes root and body prediction to minimize motion artifacts while allowing for flexible constraint conditioning. Experiments on the large-scale mocap dataset justify key design decisions and analyze how the scaling of dataset size and model size affect performance.

1. Introduction

While human motion data has always been central to games and other media, recent advances in robotics and physical AI have increased the demand for such data. In robotics, human demonstrations allow humanoids to move realistically and complete complex tasks [39, 40, 20]. Digital twins and industrial simulations need dynamic humans to populate and interact with environments. And in established domains such as game development, the growing scope of interactive experiences necessitates plausible digital humans at scale.

Obtaining high-quality 3D human motion data is challenging, though. Traditional hand animation is often tedious and requires substantial domain expertise, while studio motion capture (mocap) is expensive and requires heavy instrumentation of both the actors and environments. Teleoperation has become a popular method to collect demonstrations directly with robots [49], but this process is slow and results in awkward, unnatural behaviors. While videos offer a rich source of human motions [15], recovering high-quality 3D motions from monocular videos remains a challenging research problem.

In this work, we contend that generative models offer an alternative motion data acquisition paradigm, which can easily synthesize high-quality human motion data with precise control. For such a model to be useful in practice, it should maintain the advantages of traditional motion acquisition approaches while increasing accessibility for novice animators (*e.g.*, roboticists). In particular, an ideal model should: (1) produce high-quality motions on par with optical mocap, (2) provide a versatile and directable interface akin to hand animation, but with more intuitive controls, and (3) be able to generate a diverse corpus of motions to support a large variety of applications.

Kimodo ：扩展可控人体动作生成

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摘要

高质量的人体运动数据在机器人、仿真和娱乐领域的应用日益重要。最近的生成模型提供了潜在的数据来源，通过文本提示或姿态运动学约束等直观输入实现人体运动合成。然而，公共动作捕捉数据集的规模较小，限制了这些模型的运动质量、控制精度和泛化能力。在本工作中，我们介绍了Kimodo，一种表现力强且可控的运动学运动扩散模型，基于700小时的光学动作捕捉数据训练。我们的模型生成高质量运动，同时通过文本和一套全面的运动学约束（包括全身关键帧、稀疏关节位置/旋转、2D路径点和密集2D路径）轻松控制。这得益于精心设计的运动表示和两阶段去噪器架构，该架构将根部和身体预测分解，以最小化运动伪影，同时允许灵活的约束条件。在大规模动作捕捉数据集上的实验验证了关键设计决策，并分析了数据集规模和模型规模对性能的影响。

1. 引言

虽然人体运动数据一直是游戏和其他媒体的核心，但近期机器人和物理AI的进展加大了对这类数据的需求。在机器人领域，人类演示使仿人机器人能够逼真地移动并完成复杂任务[39, 40, 20]。数字孪生和工业模拟需要动态的人类来填充环境并与之互动。而在游戏开发等成熟领域，互动体验范围的扩大也要求大规模生成可信的数字人类。

然而，获取高质量的三维人体运动数据颇具挑战。传统手工动画往往繁琐且需要大量专业知识，而影棚动作捕捉（mocap）成本高昂，且需要对演员和环境进行大量设备布置。遥操作已成为直接通过机器人收集演示的流行方法[49]，但这一过程缓慢，且产生的行为笨拙、不自然。虽然视频提供了丰富的人体运动来源[15]，但从单目视频中恢复高质量的三维运动仍是一个具有挑战性的研究问题。

在本工作中，我们认为生成模型提供了一种替代性的运动数据获取范式，能够轻松合成高质量且可精确控制的人体运动数据。要使此类模型在实践中发挥作用，它应保留传统运动获取方法的优势，同时提高新手动画师（如机器人专家）的可及性。具体而言，理想的模型应：（1）生成与光学动作捕捉相当的高质量运动；（2）提供类似手工动画的多功能、可引导界面，但控制更直观；（3）能够生成多样化的运动语料，以支持各种应用。

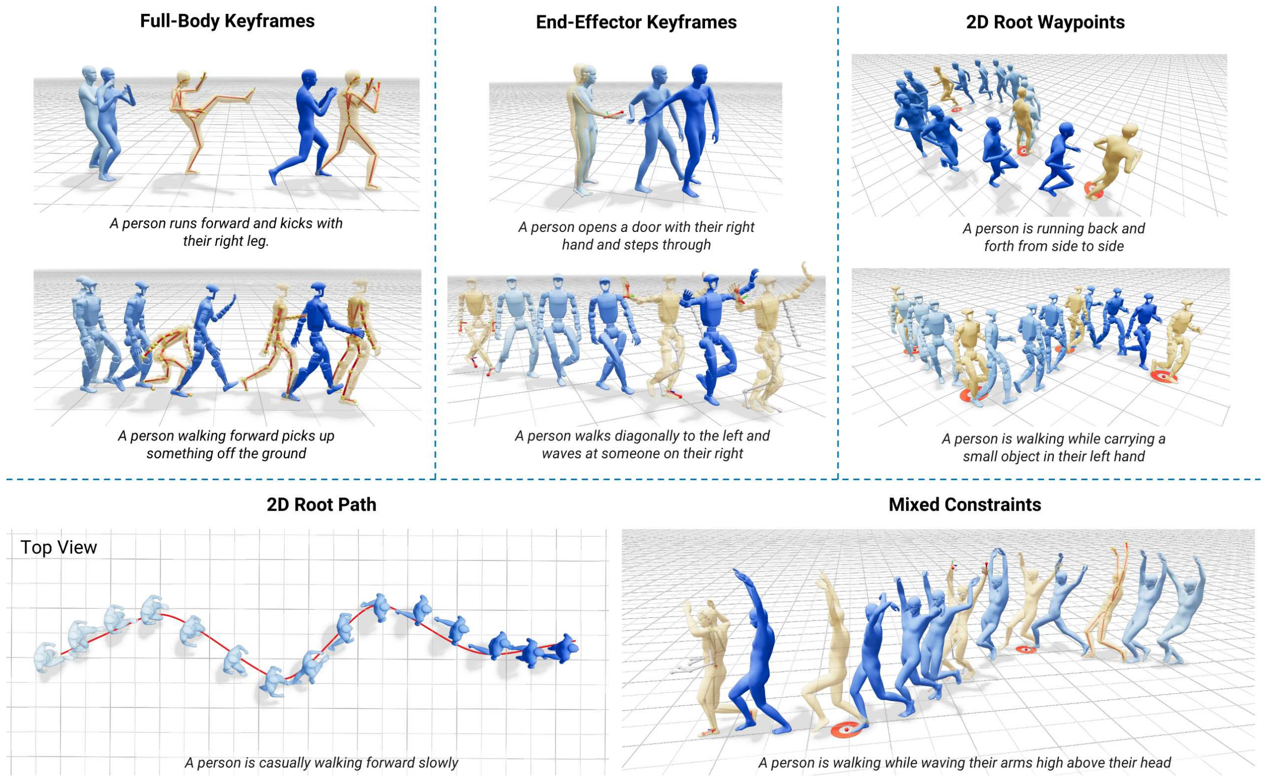

Figure 1: **Controllable Motion Generation.** Kimodo supports flexible and intuitive control for motion generation through text prompting combined with an extensive suite of kinematic constraints. By training on 700 hours of optical mocap data, the model achieves precise control accuracy for a large variety of behaviors. In each example, constrained joints are indicated with a red color, and generated poses at constrained frames are highlighted in yellow. Time progression is indicated by lighter to darker blue coloring.

Recently, rapid progress has been made in human motion generation. Modern generative models such as diffusion [42, 52], masked models [9], and tokenized transformers [50] have enabled intuitive motion generation by conditioning on text prompts. Some approaches also support *control* over generation through various kinematic constraints such as keyframe poses, 2D waypoints, and joint trajectories [9, 14, 47, 33]. These works show promising results and have inspired a wave of new research, but their motion quality, control accuracy, and generalization ability are limited by relatively small publicly available datasets [8, 28]. Furthermore, saturated benchmarks on these datasets make it difficult to differentiate which design decisions are critical to achieve effective modeling. Several works attempt to scale up human motion generation models and improve generalization by training on large volumes of motion data recovered from videos [38, 46, 19, 16], but these approaches tend to compromise motion quality due to the inherent inaccuracies in video reconstruction.

In this report, we introduce **Kimodo**, a **ki**nematic **mo**tion **d**iffusi**o**n model that trains at scale to enable intuitive authoring of high-quality motions through text prompting and an extensive set of kinematic constraints. Our model uses a carefully designed motion representation and two-stage diffusion architecture that decomposes root and body motion to accurately follow user specifications while minimizing common artifacts, such as floating and foot skating. In addition to text inputs, the model natively supports a suite of constraints on generated poses, including full-body keyframes, sparse joint positions/rotations, 2D waypoints, 2D path following, and foot contact patterns. A key component to effectively train Kimodo is the Bones Rigplay [2] dataset, a large studio mocap dataset containing 700 hours of production-quality human motion with corresponding text descriptions. This large dataset also enables a comprehensive evaluation of various design decisions on a wide range of behaviors and scenarios.

Kimodo: Scaling Controllable Human Motion Generation

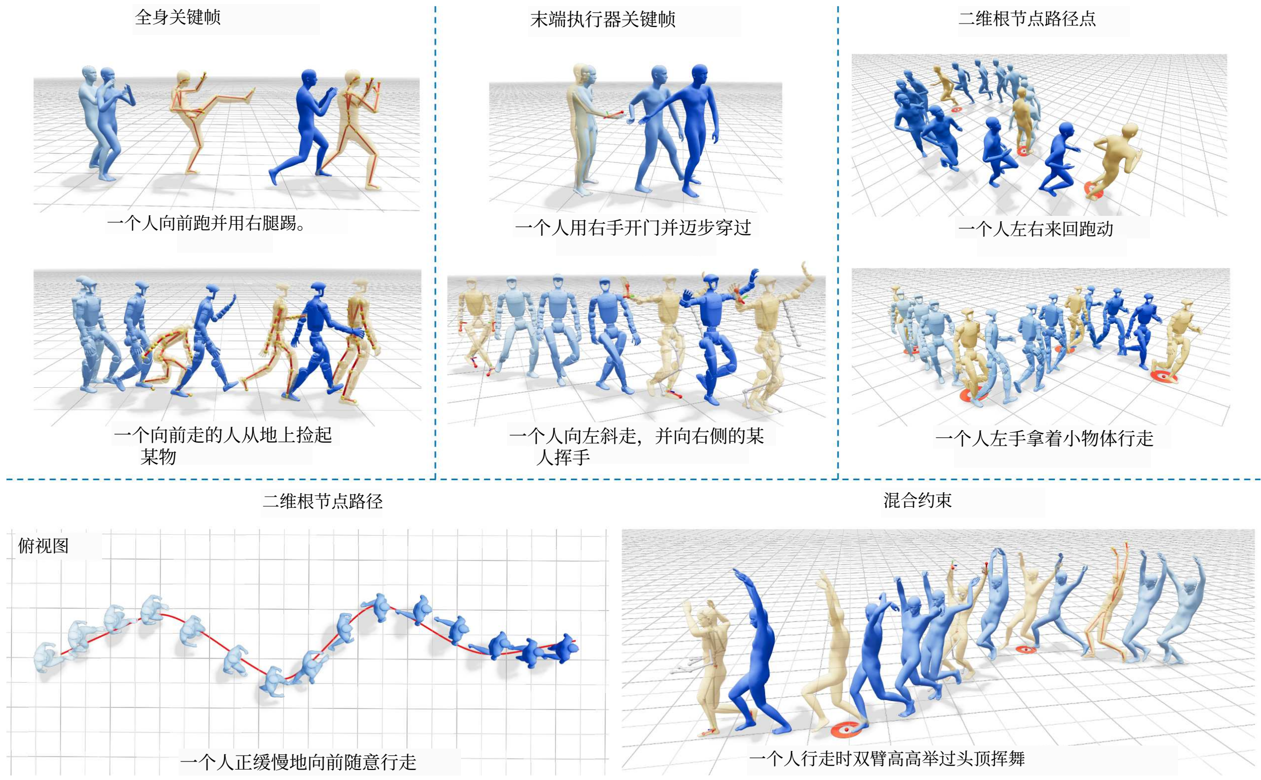

图1：可控运动生成。Kimodo通过文本提示结合广泛的运动学约束套件，支持灵活直观的运动生成控制。通过在700小时光学动作捕捉数据上训练，模型在大量行为上实现了精确的控制精度。在每个示例中，受约束的关节以红色标示，受约束帧上的生成姿态以黄色高亮。时间推进以从浅到深的蓝色着色表示。

近年来，人体运动生成取得了快速进展。现代生成模型，如扩散模型[42, 52]、掩码模型[9]和分词变换器[50]，已通过文本提示实现了直观的运动生成。一些方法还支持通过多种运动学约束（如关键帧姿态、2D路径点和关节轨迹）来控制生成[9, 14, 47, 33]。这些工作展示了有前景的结果，并激发了新一轮研究浪潮，但其运动质量、控制精度和泛化能力受限于相对较小的公开数据集[8, 28]。此外，这些数据集上饱和的基准使得难以区分哪些设计决策对实现有效建模至关重要。一些工作尝试通过从视频中恢复的大量运动数据来扩展人体运动生成模型并提高泛化能力[38, 46, 19, 16]，但这些方法往往因视频重建固有的不准确性而牺牲运动质量。

在本报告中，我们介绍Kimodo，一种运动学运动扩散模型，通过大规模训练，实现通过文本提示和广泛的运动学约束集直观创作高质量运动。我们的模型采用精心设计的运动表示和两阶段扩散架构，将根运动和身体运动分解，以准确遵循用户规格，同时最小化常见伪影，如漂浮和脚部滑动。除文本输入外，模型原生支持对生成姿态的一系列约束，包括全身关键帧、稀疏关节位置/旋转、2D路径点、2D路径跟随和脚部接触模式。有效训练Kimodo的关键组件是Bones Rigplay [2]数据集，这是一个大型工作室动作捕捉数据集，包含700小时制作级人体运动及相应文本描述。这一大型数据集还使得能够在广泛的行为和场景中对各种设计决策进行全面评估。

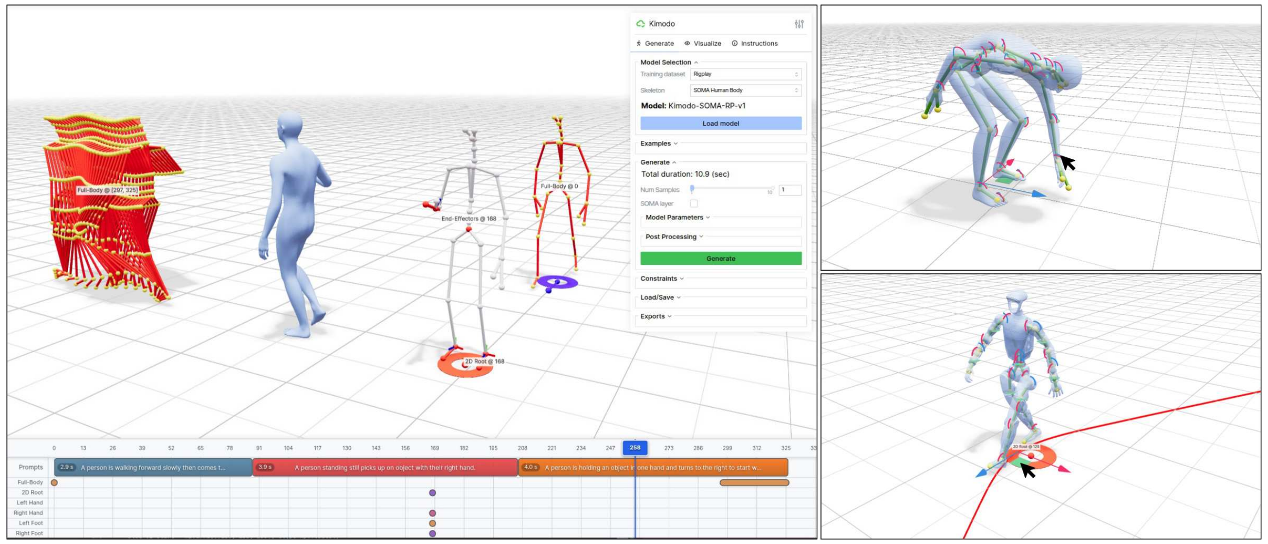

Figure 2: **Motion Authoring Demo.** (Left) Our authoring interface built with Viser [48] allows intuitive control over Kimodo for motion generation. The timeline panel allows users to specify text prompts and constraints at specific frames or intervals, which are displayed in the 3D viewer. The options panel on the right side of the interface controls various generation parameters. (Right) In editing mode, users have fine-grained control to pose and translate the character at constrained frames. Editing and generation can be done on either the SOMA body skeleton [32] or Unitree G1 robot.

To facilitate motion generation for robotics, simulation, games, and other applications, we have released model checkpoints for Kimodo trained on the SOMA body model [32] and the Unitree G1 robot [43]. Along with these models, we have included generation code and an interactive interface for motion authoring (see Fig. 2) to demonstrate the model's full capabilities. In the remainder of this report, Sec. 2 shows the key capabilities of our model, which enable practical authoring of human motions. Next, Sec. 3 describes the Bones Rigplay dataset and Sec. 4 details the core design of our two-stage transformer denoiser along with the training recipe. Sec. 5 positions our work in the context of prior systems, and lastly, Sec. 6 provides a comprehensive evaluation of key design decisions in our model and explores how scaling dataset size, model size, and batch size (GPUs) impacts model performance.

2. Key Results

Kimodo is designed for intuitive authoring of high-quality human motions. To showcase its capabilities, we developed an interactive demo with Viser [48], which allows a user to generate motions from a combination of text prompts and kinematic constraints (see Fig. 2). In this section, we detail the key features of the model within this demo interface, explore how scaling affects the model performance, and demonstrate downstream use of the model for humanoid robotics. The motions produced by our model are best viewed in the supplementary videos on the project webpage (https://research.nvidia.com/labs/sil/projects/kimodo/) or by running the demo in the released codebase (https://github.com/nv-tlabs/kimodo).

2.1. Motion Generation with Text Control

As shown in Fig. 3, motion generation with Kimodo can be easily controlled through intuitive text prompting. Our model, trained on the SOMA body skeleton [32] at 30 fps, enables generating realistic human motions for a variety of behaviors contained in the training data, including locomotion, everyday activities, dancing, actions and combat, and different motion styles. The model can handle potentially complex text descriptions that contain multiple actions performed in sequence ("A person walks forward *then* waves their arms.") or simultaneously ("A person walks forward *while* waving their arms."). Additionally, we retargeted the training

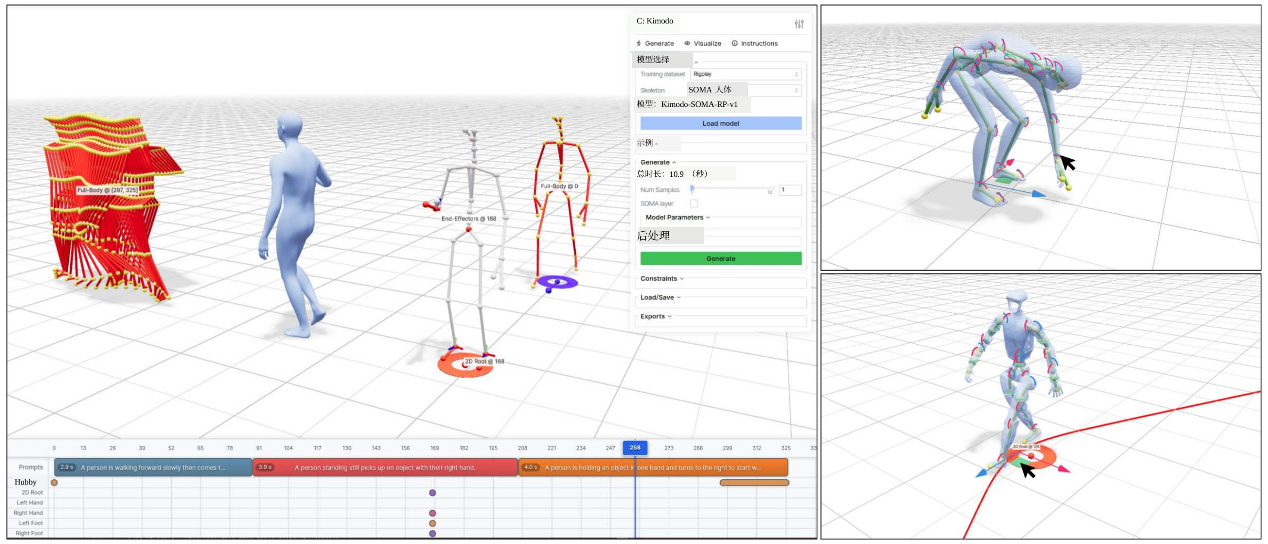

图2：运动创作演示。（左）我们使用Viser [48]构建的创作界面允许对Kimodo进行直观的运动生成控制。时间线面板允许用户在特定帧或时间间隔指定文本提示和约束，这些内容显示在3D查看器中。界面右侧的选项面板控制各种生成参数。（右）在编辑模式下，用户可以在受约束帧上对角色进行精细的姿态和位移控制。编辑和生成可在SOMA人体骨骼[32]或Unitree G1机器人上进行。

为促进机器人、仿真、游戏及其他应用的运动生成，我们发布了在SOMA人体模型[32]和Unitree G1机器人[43]上训练的Kimodo模型检查点。除这些模型外，我们还提供了生成代码和用于运动创作的交互界面（见图2），以展示模型的全部能力。在本报告的其余部分，第2节展示了我们模型的关键能力，这些能力实现了人体运动的实用创作。接下来，第3节描述Bones Rigplay数据集，第4节详述我们两阶段变换器去噪器的核心设计及训练方案。第5节将我们的工作置于先前系统的背景下，最后，第6节对我们模型中的关键设计决策进行全面评估，并探讨数据集规模、模型规模和批大小（GPU）对模型性能的影响。

2. 关键结果

Kimodo 旨在实现高质量人体动作的直观创作。为展示其能力，我们使用 Viser [48] 开发了一个交互式演示，允许用户通过文本提示和运动学约束的组合来生成动作（见图 2）。在本节中，我们详细介绍了该演示界面中模型的关键特性，探讨了规模扩展对模型性能的影响，并展示了该模型在人形机器人领域的下游应用。我们模型生成的动作最好通过项目网页上的补充视频（https://research.nvidia.com/labs/sil/projects/kimodo/）或运行发布代码库中的演示（https://github.com/nv-tlabs/kimodo）来观看。

2.1 文本控制下的运动生成

如图3所示，通过直观文本提示可轻松控制Kimodo运动生成。模型在SOMA人体骨骼[32]上以30fps训练，能生成行走、日常活动、舞蹈、动作战斗及不同风格等逼真运动。该模型支持处理含顺序多动作（如“前行后挥臂”）或同时多动作（如“边前行边挥臂”）的复杂描述。此外，利用SOMA重定向器[23]将数据映射至Unitree G1机器人，直接生成跌倒恢复及物体交互等关键技能演示。图3底行展示运动多样性：同一帧下十个不同颜色样本均响应“放下物体”提示，生成了单手/双手、高/低放及不同节奏等合理变体。

Figure 3: **Text-to-Motion Results.** (Top) Kimodo enables generating high-quality human motions for a variety of behaviors on the SOMA body skeleton. Time progression is indicated by lighter to darker blue coloring. (Middle) Motions can also be generated directly on the G1 robot to easily collect plausible demonstrations. (Bottom) The same frame is visualized from ten different generated motion samples for the same prompt, demonstrating the diversity of Kimodo outputs.

dataset to the Unitree G1 robot using the SOMA retargeter [23], allowing direct generation of kinematic robot demonstrations for important skills like recovering from stumbles and falls, and object interactions. The bottom row of Fig. 3 demonstrates the diversity of generated motions. The same frame from ten samples from the model is visualized, each with a different color. Though all samples use the same text prompt describing a put-down motion, the model generates plausible variations including one and two-handed, high and low, and variable timing.

Kimodo is trained to take a single prompt as input and expects the prompt to start with the subject, such as "A person...", "An old person...", or "A zombie...". However, our interactive authoring demo enables chaining together multiple prompts in sequence with plausible transitions. As shown in Fig. 4, using multiple prompts can be effective for performing a sequence of actions that the model may struggle with when specified as a single prompt. Please see Sec. 4.4 for technical details of multi-prompt generation.

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图3：文本到运动的结果。（顶部）Kimodo能够在SOMA人体骨骼上为各种行为生成高质量的人体运动。时间进展通过从浅蓝到深蓝的颜色表示。（中部）也可以直接在G1机器人上生成运动，以便轻松收集合理的演示。（底部）对于相同的提示，从十个不同的生成运动样本中可视化同一帧，展示了Kimodo输出的多样性。

数据集使用SOMA重定向器[23]重定向到Unitree G1机器人，从而能够直接生成关键技能（如从绊倒和跌倒中恢复以及物体交互）的运动学机器人演示。图3的底行展示了生成运动的多样性。来自模型的十个样本的同一帧被可视化，每个样本使用不同的颜色。尽管所有样本都使用相同的文本提示来描述放下物体的动作，但模型生成了合理的变体，包括单手和双手、高放和低放以及不同的时间节奏。

Kimodo被训练为接受单个提示作为输入，并期望提示以主语开头，例如"一个人..."、"一位老人..."或"一个僵尸..."。然而，我们的交互式创作演示支持将多个提示按顺序链接起来，并带有合理的过渡。如图4所示，使用多个提示可以有效地执行模型在单个提示下可能难以处理的动作序列。多提示生成的技术细节请参见第4.4节。

Kimodo is designed as an offline motion authoring model, and is best suited for generating one or more motions in a batched fashion. On an NVIDIA RTX 3090, generating a motion from a single prompt takes anywhere from 2 to 5 sec, depending on the specified duration. The model is trained on a maximum of 10 sec sequences.

2.2. Kinematic Control with Pose Constraints

Precise control can be achieved through kinematic constraints on generated poses. As shown in Fig. 1, Kimodo supports a wide range of constraint types. *Full-body keyframes* constrain all the joint positions of the character at specific frames. *End-effector keyframes* specify the position and rotation of one or more hand or foot joints. For 2D root constraints, *waypoints* specify sparse targets on the ground for the character to hit, or a full dense *path* can be used to constrain an interval of motion. Constraints can be mixed arbitrarily to achieve desired motions.

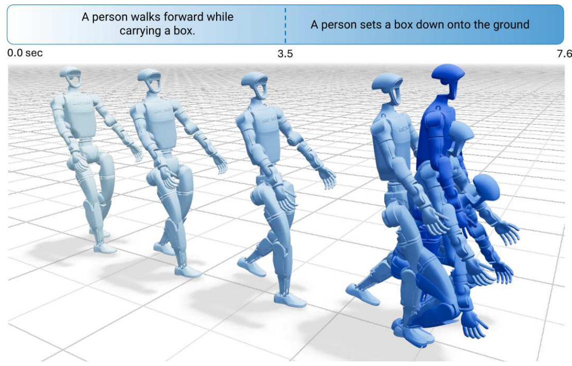

Figure 4: **Multi-Prompt Generation.** Longer motion sequences can be generated from multiple prompts with the demo's timeline interface. Motions are generated sequentially with constraints between them for continuity.

Kinematic control enables several useful motion authoring applications. Animators can use Kimodo to generate plausible transitions between existing mocap clips through in-betweening, with full-body constraints placed at the start and end of the generated sequence. For characters to realistically move about an environment, 2D paths or waypoints can be specified automatically using traditional animation tools like navigation meshes. For object interactions, users or automated planners can place constraints, such as hand positions, on objects to encourage Kimodo to generate pick and place motions that are plausible for a specific object size. To generate large scale motion datasets, constraints can be randomized to ensure diversity. For example, for creating a locomotion move set, a 2D root waypoint can be placed at varying angles from the starting location to synthesize locomotion in all different directions.

We evaluated Kimodo trained for the SOMA skeleton on a diverse test suite of constraint-conditioned motion generation cases (see Sec. 6.1 for details). On average, the model achieves 3.21 cm joint errors for full-body keyframes, 3.63 cm joint position errors for end-effector constraints, 6.88 deg joint rotation errors for end-effectors, and 3.63 cm root errors across waypoints and paths. Given such precise model outputs, light post-processing can be applied on generated motions to ensure they *exactly* hit user constraints without severely degrading motion quality. Note that, while we use the same evaluation protocol, these numbers are not comparable to those in Sec. 6, where models are trained on a different character skeleton at 20 fps.

2.3. Scaling Behavior

A key component of Kimodo's success is scaling along several axes. To demonstrate this, we evaluate the model while varying data size, model size, and batch size (number of GPUs). We summarize key results in this section, while full details are provided in Sec. 6.3.

Fig. 5 plots key metrics affected by each scaling axis. Increasing dataset size is particularly helpful for the model to see a larger variety of motions during training, which improves its ability to follow constraints and decreases errors across all constraint types. Increasing model size greatly improves text-following capabilities, as indicated by R-precision, along with motion quality, as indicated by FID. Finally, increasing the number of GPUs available for training, thereby increasing batch size, allows for further improvements particularly in text-following.

Kimodo 被设计为一种离线动作生成模型，最适合以批量方式生成一个或多个动作。在 NVIDIA RTX 3090 上，根据指定的时长，从单个提示生成一个动作需要 2 到 5 秒。该模型在最长 10 秒的序列上进行训练。

2.2. 基于姿态约束的运动学控制

通过对生成姿态施加运动学约束，可以实现精确控制。如图1所示，Kimodo支持多种约束类型。全身关键帧在特定帧上约束角色的所有关节位置。末端执行器关键帧指定一个或多个手部或脚部关节的位置和旋转。对于2D根约束，路径点指定角色在地面上需要到达的稀疏目标，或者可以使用完整的密集路径来约束一段运动区间。这些约束可以任意组合，以实现所需的运动效果。

图4：多提示生成。通过演示时间线界面，可从多个提示生成长动作序列。动作按序生成并施加约束以确保连续性。

运动学控制支持多种实用的运动创作应用。动画师可以使用Kimodo在现有动作捕捉片段之间生成合理的过渡，通过插值方式，在生成序列的起始和结束位置放置全身约束。为了让角色在环境中真实地移动，可以使用传统动画工具（如导航网格）自动指定2D路径或路径点。对于物体交互，用户或自动规划器可以在物体上放置约束（如手部位置），以促使Kimodo生成对特定物体尺寸合理的抓取和放置动作。为了生成大规模运动数据集，可以随机化约束以确保多样性。例如，在创建行走动作集时，可以在起始位置的不同角度放置2D根路径点，以合成各个方向的行走动作。

我们在针对SOMA骨架训练的Kimodo模型上，对多样化的约束条件运动生成测试集进行了评估（详见第6.1节）。平均而言，模型在全身关键帧上的关节误差为3.21厘米，末端执行器约束的关节位置误差为3.63厘米，末端执行器的关节旋转误差为6.88度，路径点和路径上的根节点误差为3.63厘米。鉴于模型输出的精确性，可以对生成的运动进行轻量级后处理，以确保其精确满足用户约束，而不会严重降低运动质量。需要注意的是，虽然我们使用了相同的评估协议，但这些数字与第6节中的结果不可直接比较，因为第6节中的模型是在不同角色骨架、20帧每秒的条件下训练的。

2.3. 扩展行为

Kimodo成功的关键在于沿多个轴进行扩展。为证明这一点，我们在改变数据规模、模型规模和批处理规模（GPU数量）的情况下评估模型。本节总结关键结果，完整细节见第6.3节。

图5绘制了受每个扩展轴影响的关键指标。增加数据集规模特别有助于模型在训练期间看到更多样化的动作，从而提高其遵循约束的能力，并减少所有约束类型的误差。增加模型规模大幅提升文本跟随能力（如R-precision所示）以及动作质量（如FID所示）。最后，增加可用于训练的GPU数量，从而增大批处理规模，可进一步改进，尤其是在文本跟随方面。

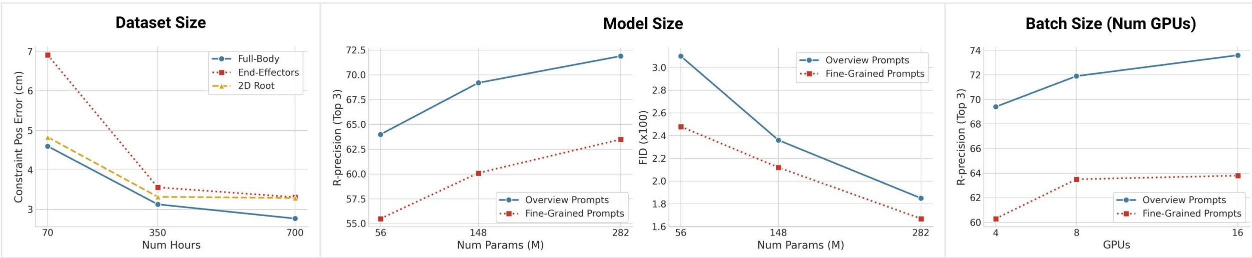

Figure 5: **Scaling Results.** Scaling dataset size, model size, and batch size improves controllability and motion quality. Increased dataset size results in greatly improved constraint following, while model size and batch size are particularly helpful for text following (R-precision) and motion quality (FID). See Tab. 2 for full results.

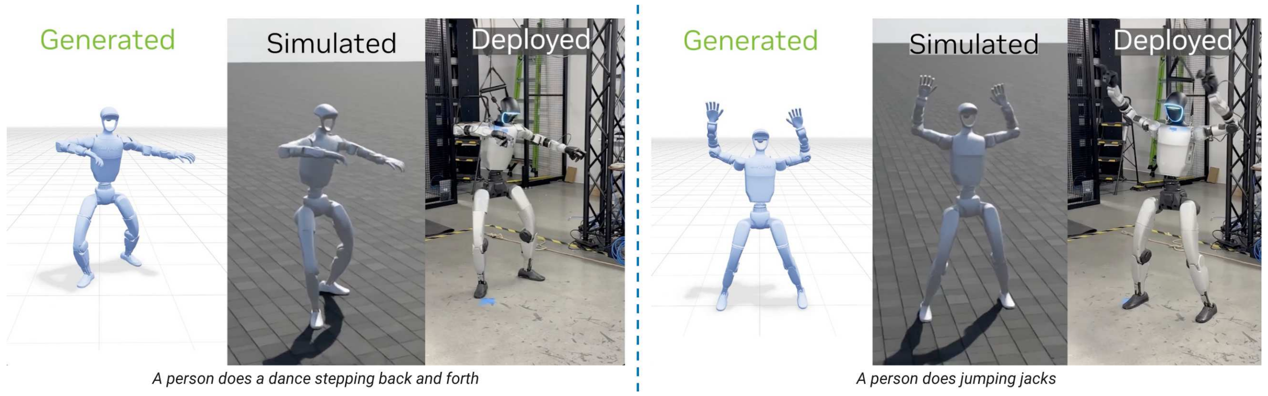

Figure 6: **Generating Robot Demonstrations.** In these results, Kimodo is used to generate demonstration data directly on the G1 robot, which is then tracked by a physics-based humanoid policy trained with ProtoMotions [41] and deployed to a real-world robot.

2.4. Application: Demonstration Data for Humanoid Robots

As shown in Fig. 6, Kimodo trained on G1 can directly generate demonstrations for robotics applications. Unlike traditional mocap or teleoperation, acquiring these motions takes a few seconds with a simple text prompt as input. Larger demonstration datasets can be created with batched generation and by employing constraints to ensure diversity of motion variations.

3. Large-Scale Motion Capture Dataset

For training and evaluation, we leverage Bones Rigplay [2], a large-scale optical motion capture dataset containing 700 hours of motions from 170 human subjects with a roughly equal number of male and female participants. The dataset contains thousands of unique actions covering a range of locomotion, gestures, everyday activities, common object interactions, videogame combat, dancing, athletics, and more. Many actions are also performed in different styles including tired, angry, happy, sad, scared, drunk, injured, stealthy, old, and childlike. Actions are performed by multiple subjects across multiple takes, providing a rich diversity of performers, semantics, and motion variability. As shown in Fig. 7, each clip in the dataset is labeled with an *overview* text description of the entire motion at a high level. Additionally, the clip is broken down into more *fine-grained* atomic action sub-clips, each containing a text description of the contained action.

The dataset contains body-only motions (i.e., no finger motions) that have been retargeted to a uniform-proportion 27-joint skeleton to be standardized across different performers. We use this “native” 27-joint skeleton for quantitative experiments presented in Sec. 6, however, we also retarget the dataset to other skeletons to train variations of our model, including SOMA [32], the Unitree G1 Robot [43], and SMPL-X [18, 24].

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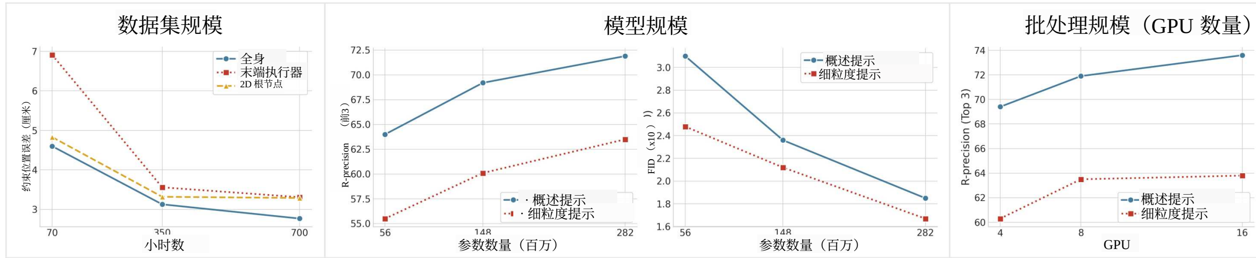

图5：扩展结果。扩展数据集规模、模型规模和批次大小可提高可控性和动作质量。数据集规模的增加显著改善了约束遵循，而模型规模和批次大小对文本遵循（R-精度）和动作质量（FID）尤为有益。完整结果见表2。

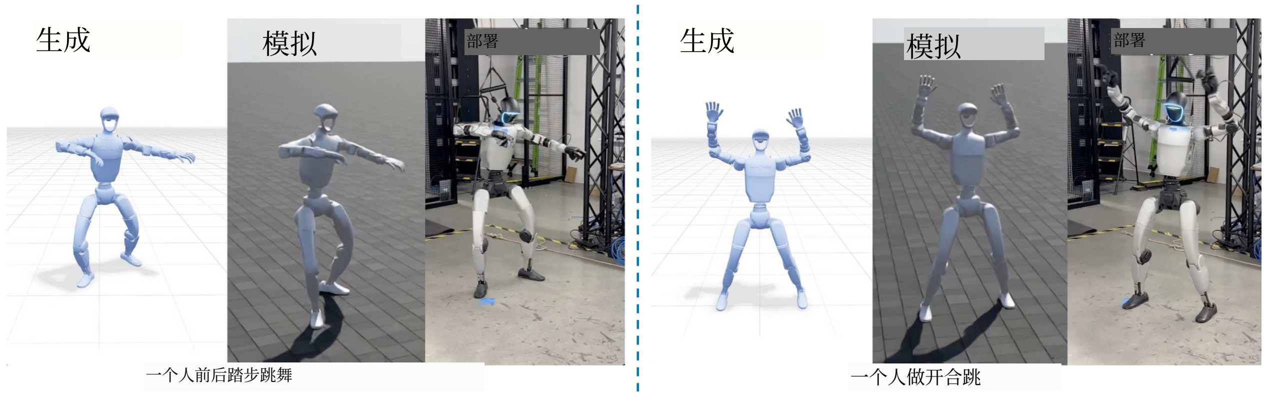

图6：生成机器人演示。在这些结果中，Kimodo被用于直接在G1机器人上生成演示数据，随后由基于物理的人形机器人策略跟踪，该策略使用ProtoMo-tions [41]训练并部署到真实机器人上。

2.4. 应用：人形机器人演示数据

如图6所示，在G1上训练的Kimodo可直接为机器人应用生成演示。与传统动作捕捉或遥操作不同，通过简单的文本提示输入，仅需几秒即可获取这些动作。通过批量生成并施加约束以确保动作变体的多样性，可以创建更大的演示数据集。

3. 大规模动作捕捉数据集

为了训练和评估，我们利用了Bones Rigplay [2]，这是一个大规模光学动作捕捉数据集，包含来自170名受试者的700小时动作，男女参与者数量大致相等。该数据集包含数千种独特动作，涵盖运动、手势、日常活动、常见物体交互、电子游戏战斗、舞蹈、田径等。许多动作还以不同风格执行，包括疲惫、愤怒、快乐、悲伤、恐惧、醉酒、受伤、隐秘、年老和孩童般。动作由多名受试者在多次拍摄中执行，提供了丰富的表演者、语义和动作变异性。如图7所示，数据集中的每个片段都标注了整体动作的高层概述文本描述。此外，该片段被细分为更细粒度的原子动作子片段，每个子片段包含所包含动作的文本描述。

该数据集包含仅身体动作（即无手指动作），已重定向到统一比例的27关节骨架，以便在不同表演者之间标准化。我们在第6节中呈现的定量实验中使用这种“原生”27关节骨架，但我们也将该数据集重定向到其他骨架以训练我们模型的变体，包括SOMA [32]、Unitree G1机器人[43]和SMPL-X [18, 24]。

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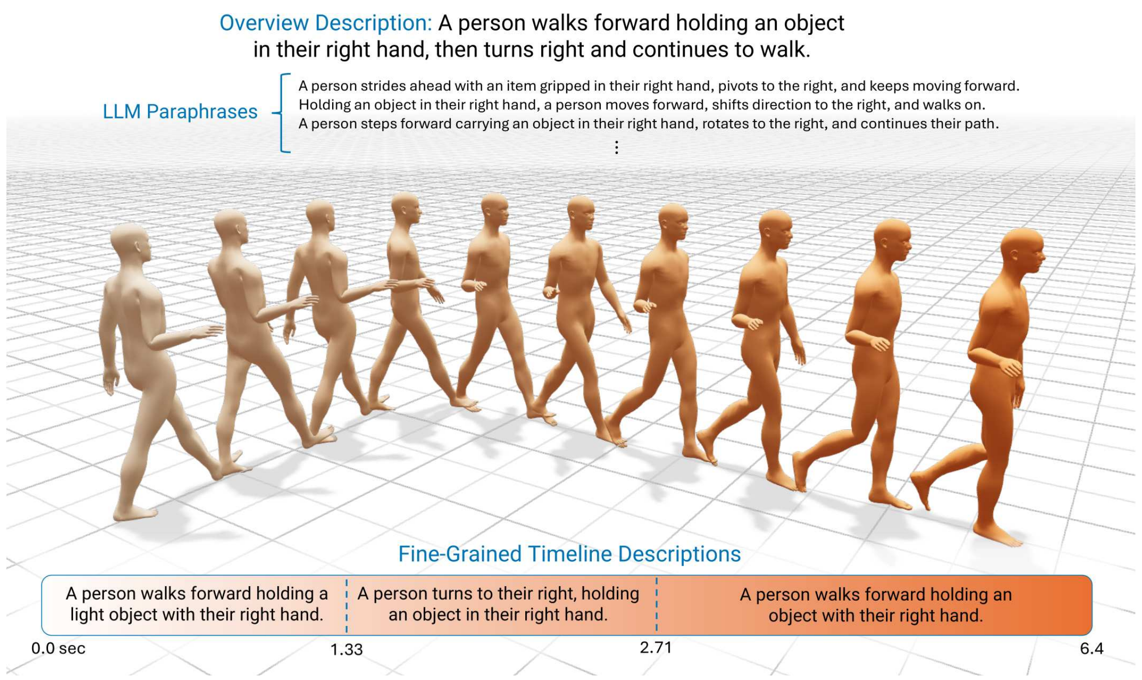

Figure 7: **Example Motion Data.** Each mocap sequence in our training dataset includes a high-level overview description, fine-grained descriptions of sub-clips containing atomic actions, and augmented LLM paraphrases.

Augmentations. For training the denoising model introduced in Sec. 4.2, we apply augmentations to both the text and motions in the dataset. For text, we use an LLM (Qwen3-32B [37]) to paraphrase the motion descriptions into a consistent prompt structure (always starting with "A [subject]...") with various levels of detail. To improve handling complex prompts that contain a composition of multiple actions, we also augment the motion in the dataset by stitching together random pairs of motion clips. To ensure natural transitions between the stitched clips, we use our diffusion model trained on the non-augmented dataset to generate short transition motions between the clips.

During training, we randomly sample from the available data and augmentation variations according to a pre-specified distribution. In particular, we train on a combination of full motion clips, single or combined action sub-clips, augmented stitched motion clips, original text descriptions, and augmented LLM paraphrases.

4. Method: Kimodo

Our model is designed to provide users with an intuitive and versatile interface for authoring high-quality motions. The core component of our framework is an explicit motion diffusion model [42, 52], which has been shown to effectively capture the complex distribution of text and motion. It also enables intuitive kinematic controls through simple conditioning mechanisms, such as direct imputation of pose features [33, 34].

Background. Explicit motion diffusion [42, 52] applies ideas from earlier image diffusion models [11], but operates directly on body pose features. Given a clean motion $\mathbf{x}_0$, the forward diffusion process is defined as a Gaussian process that adds noise to the poses until they are approximately $\mathcal{N}(\mathbf{0}, \mathbf{I})$. To generate a motion, the reverse process is used to iteratively denoise a sequence of noisy human poses $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. To enable this reverse process, a denoiser $\mathcal{D}_\theta(\mathbf{x}_t, C, t)$, which takes a noisy motion $\mathbf{x}_t$, conditioning signals C, and the current denoising step $t \in \{1, \dots, T\}$ as input, is trained to output a prediction of the clean motion $\hat{\mathbf{x}}_0$. The predicted clean motion is then re-noised to obtain an estimate of $\mathbf{x}_{t-1}$, which serves as input to the next denoising step in the reverse diffusion process. The conditioning signals may include text or other constraints on the motion.

The key design decisions to enable high-quality motion and control are the motion representation (Sec. 4.1)

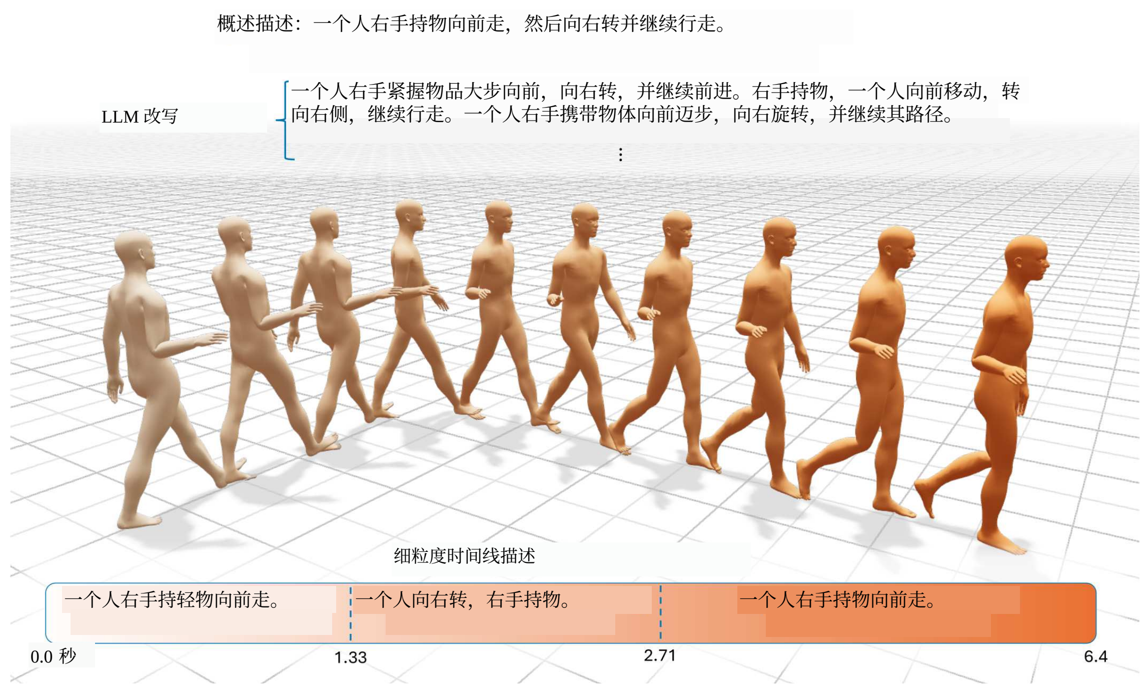

图7：示例动作数据。我们训练数据集中的每个动作捕捉序列都包含高层概述描述、包含原子动作的子片段的细粒度描述，以及增强的LLM改写。

增强。为了训练第4.2节中引入的去噪模型，我们对数据集中的文本和动作都应用增强。对于文本，我们使用LLM（Qwen3-32B [37]）将动作描述改写为一致的提示结构（始终以“A [subject]...”开头），具有不同详细程度。为了改进对包含多个动作组合的复杂提示的处理，我们还通过拼接随机成对的动作片段来增强数据集中的动作。为了确保拼接片段之间的自然过渡，我们使用在非增强数据集上训练的扩散模型生成片段之间的短过渡动作。

在训练期间，我们根据预定的分布从可用数据和增强变体中随机采样。具体来说，我们训练于完整动作片段、单个或组合动作子片段、增强拼接动作片段、原始文本描述和增强LLM改写的组合。

4. 方法：Kimodo

我们的模型旨在为用户提供直观且多功能的界面，用于创作高质量的动作。我们框架的核心组件是一个显式动作扩散模型 [42, 52]，该模型已被证明能有效捕捉文本和动作的复杂分布。它还通过简单的条件机制（如直接插补姿态特征 [33, 34]）实现直观的运动学控制。

背景。显式动作扩散 [42, 52] 借鉴了早期图像扩散模型 [11] 的思想，但直接作用于身体姿态特征。给定一个干净的动作 $\mathbf{x}_0$，前向扩散过程被定义为一个高斯过程，向姿态添加噪声，直到它们近似为 $\mathcal{N}(\mathbf{0}, \mathbf{I})$。为了生成动作，使用反向过程对一系列带噪声的人体姿态 $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 进行迭代去噪。为了实现这一反向过程，一个去噪器 $\mathcal{D}_\theta(\mathbf{x}_t, C, t)$，以带噪声的动作 $\mathbf{x}_t$、条件信号 C 和当前去噪步骤 $t \in \{1, \dots, T\}$ 作为输入，被训练为输出干净动作的预测 $\hat{\mathbf{x}}_0$。预测的干净动作随后被重新加噪，以获得 $\mathbf{x}_{t-1}$ 的估计，该估计作为反向扩散过程中下一步去噪的输入。条件信号可能包括文本或其他对动作的约束。

实现高质量动作和控制的关键设计决策是动作表示（第 4.1 节）

and the denoiser architecture (Sec. 4.2). In the following sections, we describe these components of our system, along with the training recipe (Sec. 4.3) and generation process (Sec. 4.4).

4.1. Motion Representation

A motion sequence is represented as $\mathbf{x} = [\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^N]$, where N is the number of frames. We assume the motion is standardized such that the root at the first frame is above the origin. For a skeleton with J joints, each pose in the motion is represented by a vector of features $[\mathbf{r}^p, \mathbf{r}^a, \mathbf{j}^p, \mathbf{j}^v, \mathbf{j}^a, \mathbf{f}]$ consisting of:

- $\mathbf{r}^p \in \mathbb{R}^3$: **smoothed global root position**. The root motion is computed by taking the pelvis position and heavily smoothing its 2D horizontal components (x, z), while keeping the y height unchanged.
- $\mathbf{r}^a \in \mathbb{R}^2$: **global root heading direction**. This is represented as $[\cos(\psi), \sin(\psi)]$ with the heading angle ψ. The heading angle is computed based on the projection of the cross product $\mathbf{e}_y \times \mathbf{v}_{\text{hips}}$ onto the xz (ground) plane, where $\mathbf{e}_y$ is the unit up vector and $\mathbf{v}_{\text{hips}}$ is the vector from left to right hip joints.
- $\mathbf{j}^p \in \mathbb{R}^{3J}$: **joint positions**. The 2D horizontal components (x, z) of each joint are represented relative to the smoothed root position, while the y height is global. Note that these joint positions are *not* rotated to be relative to the root heading direction.
- $\mathbf{j}^v \in \mathbb{R}^{3J}$: **global joint velocities**. The joint velocities are computed from the *global* joint positions, rather than the (partially) root-relative positions $\mathbf{j}^p$.
- $\mathbf{j}^a \in \mathbb{R}^{6J}$: **global joint angles** encoded using the 6D representation [55].
- $\mathbf{f} \in \{0, 1\}^4$: **foot contacts** boolean flags for the [left heel, left toe, right heel, right toe].

Considering the decomposed architecture introduced later, it is helpful to think of each pose as containing a global root component $\mathbf{r}^{\text{glob}} = [\mathbf{r}^p, \mathbf{r}^a]$ and a body component $\mathbf{b} = [\mathbf{j}^p, \mathbf{j}^v, \mathbf{j}^a, \mathbf{f}]$.

Discussion. This motion representation is carefully designed for motion quality and to be amenable for conditioning through direct imputation (*i.e.*, overwriting) of input pose features during denoising (see Sec. 4.2). First, our mostly global representation allows for sparse constraints on the root, joint positions, and joint rotations throughout a pose sequence. This is challenging for other common representations [8] that are purely local/velocity-based, since they require integration to determine global positions/rotations at a specific frame and are therefore better suited for temporally dense constraints [22]. Second, our joint position representation is not canonicalized with respect to the root heading in each frame. In prior work, joint positions are commonly canonicalized with respect to the root heading, but this design decision can lead to sudden changes in the local joint position representation due to abrupt flips in heading direction, such as during somersaults and cartwheels. These discontinuities can lead to training instabilities and compromise motion quality. Third, using a global joint rotation representation enables users to directly specify sparse joint rotation constraints in world space (through imputation), which is rarely supported in prior systems. While this is a common requirement for animation applications, it is difficult to achieve if joint rotations are represented relative to their parent in the kinematic chain (*e.g.*, in the SMPL body model [18]), since forward kinematics through the entire chain is required to recover a joint's global orientation. Lastly, compared to directly using the pelvis position projected to the ground as the root, our smoothed root position better emulates the smooth curves and straight lines that users are likely to specify as

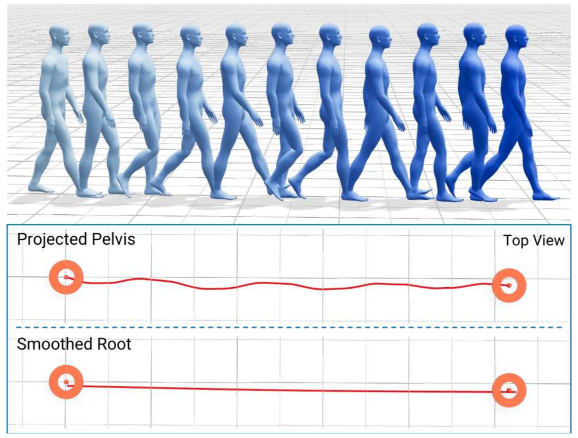

Figure 8: **Smoothed Root Representation.** For a simple walking motion, the projected pelvis path captures the sway of the hips, while the smoothed root is nearly a straight line. This smoothed trajectory offers a stable frame of reference for representing joint positions, and emulates paths drawn in practical animation tools.

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和去噪器架构（第4.2节）。下文描述这些组件、训练方案（第4.3节）及生成过程（第4.4节）。

4.1. 动作表示

动作序列表示为 $\mathbf{x} = [\mathbf{x}^1, \mathbf{x}^2, \ldots, \mathbf{x}^N]$ ，其中 N 是帧数。我们假设动作已标准化，使得第一帧的根节点位于原点上方。对于具有 J 个关节的骨架，动作中的每个姿态由特征向量 $[\mathbf{r}^p, \mathbf{r}^a, \mathbf{j}^p, \mathbf{j}^v, \mathbf{j}^a, \mathbf{f}]$ 表示，该向量包含：

- $\mathbf{r}^p \in \mathbb{R}^3$ ：平滑后的全局根位置。根运动通过取骨盆位置并对其二维水平分量 (x, z) 进行重度平滑计算，同时保持 y 高度不变。
- $\mathbf{r}^a \in \mathbb{R}^2$ ：全局根朝向，表示为 $[\cos(\psi), \sin(\psi)]$ 。朝向角 ψ 基于叉积 $\mathbf{e}_y \times \mathbf{v}_{\text{hips}}$ 在 xz 平面的投影计算，其中 $\mathbf{e}_y$ 为单位向上向量， $\mathbf{v}_{\text{hips}}$ 为从左髋到右髋的向量。
- $\mathbf{j}^p \in \mathbb{R}^{3J}$ ：关节位置。每个关节的二维水平分量 (x, z) 相对于平滑后的根位置表示，而 y 高度是全局的。注意，这些关节位置不旋转为相对于根朝向方向。
- $\mathbf{j}^v \in \mathbb{R}^{3J}$ ：全局关节速度。关节速度由全局关节位置计算，而非（部分）根相对位置 $\mathbf{j}^p$ 。
- $\mathbf{j}^a \in \mathbb{R}^{6J}$ ：使用6D 表示[55] 编码的全局关节角度。
- $\mathbf{f} \in \{0, 1\}^4$ ：脚部接触布尔标志，对应[左脚跟，左脚趾，右脚跟，右脚趾]。

考虑到后面介绍的分解架构，将每个姿态视为包含全局根组件 $\mathbf{r}^{\text{glob}} = [\mathbf{r}^p, \mathbf{r}^a]$ 和身体组件 $\mathbf{b} = [\mathbf{j}^p, \mathbf{j}^v, \mathbf{j}^a, \mathbf{f}]$ 是有帮助的。

讨论。这种动作表示经过精心设计，以保证动作质量，并便于在去噪过程中通过直接插补（即覆盖）输入姿态特征进行条件控制（见第4.2节）。首先，我们主要采用全局表示，使根节点、关节位置及旋转的稀疏约束成为可能。这克服了纯局部/速度表示[8]需积分求全局位姿的局限，后者更适于密集约束[22]。其次，我们的关节位置未针对每帧根节点朝向规范化。先前工作常相对根节点规范化，但易因朝向突变（如翻筋斗）导致不连续，引发训练不稳定并损害运动质量。第三，全局关节旋转表示支持用户直接在世界空间指定稀疏旋转约束（通过插补），而基于父节点的表示（如SMPL[18]）因需正向运动学恢复全局方向难以实现。最后，相比直接将骨盆投影至地面，我们的平滑根位置更好地模拟了用户指定的平滑曲线和直线。

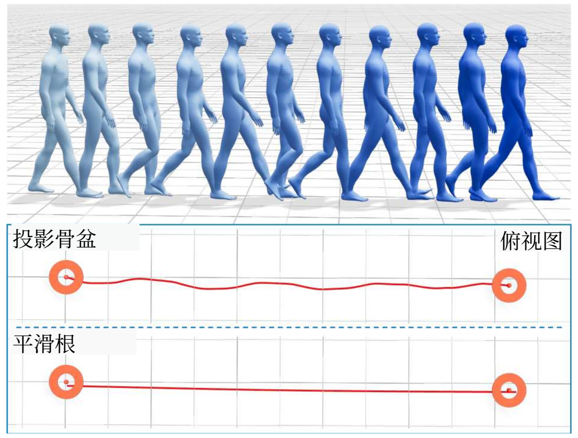

图8：平滑根表示。对于简单的行走运动，投影骨盆路径捕捉了髋部的摆动，而平滑根几乎是一条直线。这条平滑轨迹为表示关节点位置提供了稳定的参考框架，并模拟了实际动画工具中绘制的路径。

如果关节旋转相对于运动链中的父节点来表示（例如，在SMPL人体模型[18]中），则难以实现，因为需要通过整个链的正向运动学来恢复关节的全局方向。最后，与直接将骨盆位置投影到地面作为根节点相比，我们的平滑根位置更好地模拟了用户可能指定的平滑曲线和直线。

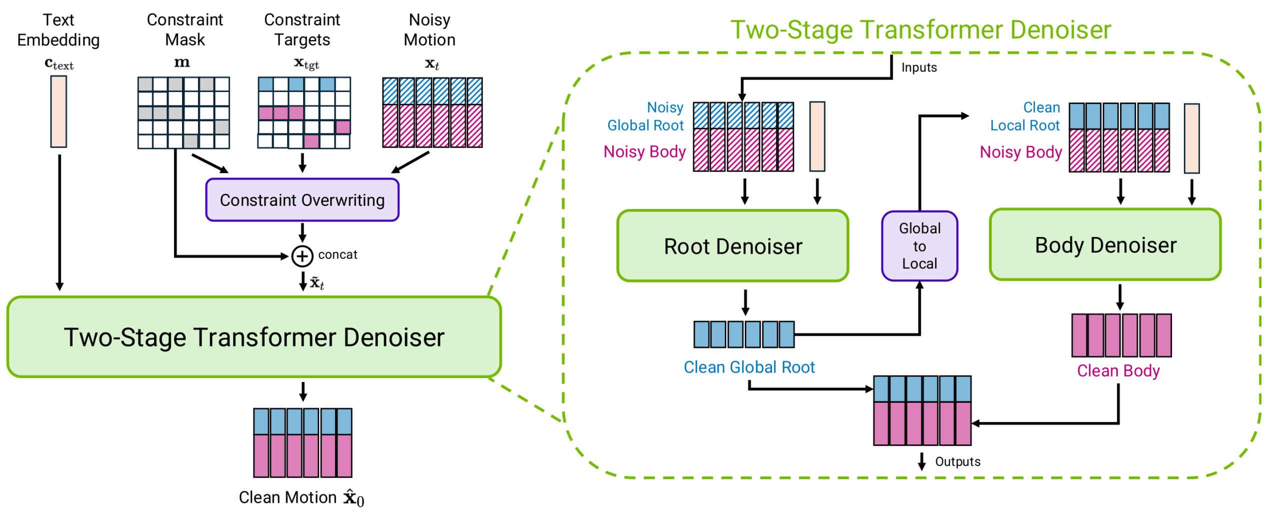

Figure 9: **Denoiser Architecture.** (Left) Kimodo predicts clean motion given a noisy motion, pose constraints, and a text embedding. Specified pose constraints directly overwrite the noisy motion before it is given to the denoiser. (Right) The two-stage denoiser decomposes root and body motion prediction. The root denoiser first predicts the global root motion, which is transformed into a local representation as input to the body denoiser. The final output of the denoising step is the concatenation of the outputs from the two stages.

2D waypoint and path constraints (Fig. 8). This smoothed root representation enables our model to generate motions that follow smooth target paths, while also providing flexibility for the pelvis joint to move naturally around the smoothed path.

4.2. Two-Stage Transformer Denoiser

Given a noisy motion $\mathbf{x}_t$ at denoising step t, the denoiser predicts the clean motion $\hat{\mathbf{x}}_0$ using a two-stage transformer architecture shown in Fig. 9. Denoising is optionally conditioned on a text prompt and/or constraints.

Inputs. The input noisy motion $\mathbf{x}_t$ to the transformer is treated as a sequence of pose tokens. Kinematic constraints are specified as partial pose features in the same representation as the input motion. In particular, the target pose features are specified by a (usually sparse) target motion $\mathbf{x}_{\text{tgt}}$ along with a binary control mask $\mathbf{m}$, which indicates which features are constrained. We impute (*i.e.*, overwrite) the noisy motion with the target pose features according to the control mask as $\tilde{\mathbf{x}}_t = \mathbf{m} \odot \mathbf{x}_{\text{tgt}} + (1 - \mathbf{m}) \odot \mathbf{x}_t$, where $\odot$ is the element-wise product [33]. Finally, we concatenate the control mask with the imputed motion along the feature dimension to produce the final input tokens $\mathbf{x}_{\text{in}} = [\tilde{\mathbf{x}}_t; \mathbf{m}]$. When no constraints are specified, the mask is simply all zeros and $\tilde{\mathbf{x}}_t = \mathbf{x}_t$.

The denoiser prediction is $\hat{\mathbf{x}}_0 = \mathcal{D}_\theta(\mathbf{x}_{\text{in}}, C, t)$ where $C = \{\mathbf{c}_{\text{text}}, \mathbf{c}_{\text{dir}}, \mathbf{c}_{\text{extra}}\}$ is the set of additional conditioning tokens. $\mathbf{c}_{\text{text}} \in \mathbb{R}^{4096}$ is the LLM2Vec embedding of the input text prompt [1], which we found to outperform common alternatives like CLIP [29] and T5 [30] in early experiments. $\mathbf{c}_{\text{dir}} \in \mathbb{R}^2$ is the desired heading direction of the first frame. Lastly, $\mathbf{c}_{\text{extra}} \in \mathbb{R}^{P \times 4096}$ contains P extra all-zero tokens. While these extra tokens cannot exactly be considered "register" tokens [4], since they are not learned, they achieve a similar effect of enhancing the representational capacity of the model as shown in our experiments (Sec. 6). In practice, we use $P = 49$. All pose and conditioning tokens are embedded to the same dimensionality and added to a sinusoidal positional encoding before going into the transformer.

Architecture. As shown in Fig. 9, the denoiser is decomposed into two transformer encoders: one for predicting root motion and the other for body motion. The root denoiser first predicts the global root motion $\hat{\mathbf{r}}_0^{\text{glob}}$. Despite only predicting the root, this denoiser is conditioned on the full noisy motion $\mathbf{x}_{\text{in}}$, such that the output can be

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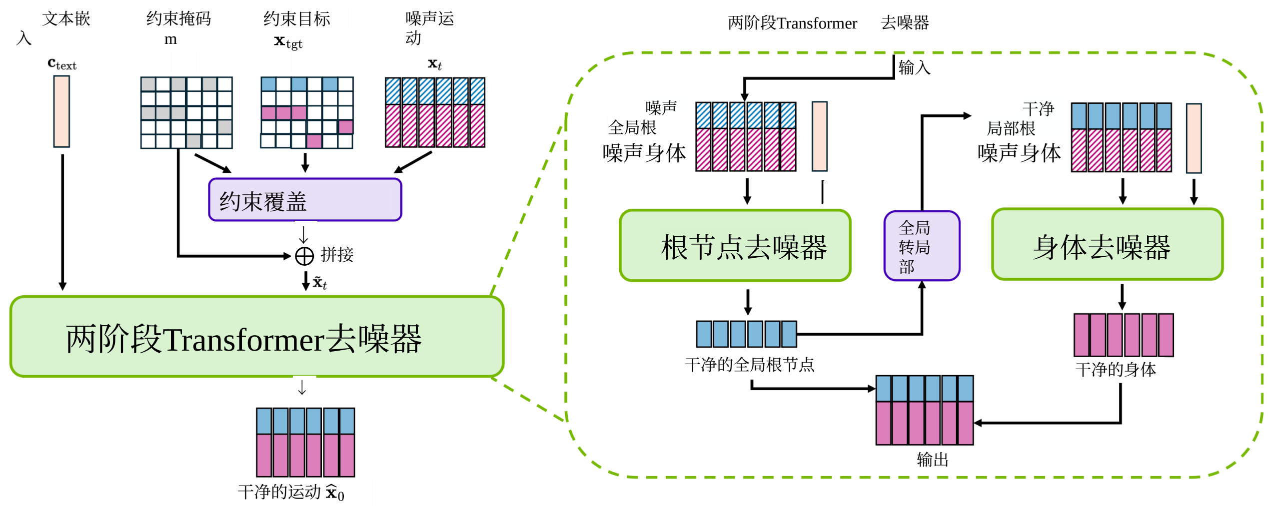

图9：去噪器架构。（左）Kimodo根据带噪声的运动、姿态约束和文本嵌入预测干净的运动。指定的姿态约束在输入去噪器之前直接覆盖带噪声的运动。（右）两阶段去噪器将根节点和身体运动预测分解。根节点去噪器首先预测全局根节点运动，该运动被转换为局部表示作为身体去噪器的输入。去噪步骤的最终输出是两个阶段输出的拼接。

2D路径点和路径约束（图8）。这种平滑的根节点表示使我们的模型能够生成遵循平滑目标路径的运动，同时为骨盆关节在平滑路径周围自然移动提供灵活性。

4.2. 两阶段Transformer 去噪器

给定去噪步骤 t 中的带噪运动 $\mathbf{x}_t$，去噪器使用图9所示的两阶段Transformer架构预测干净运动 $\hat{\mathbf{x}}_0$。去噪可选地以文本提示和/或约束为条件。

输入。输入到Transformer的带噪运动 $\mathbf{x}_t$ 被视为一系列姿态令牌。运动学约束以与输入运动相同表示的局部姿态特征指定。具体而言，目标姿态特征由（通常稀疏的）目标运动 $\mathbf{x}_{\text{tgt}}$ 和二进制控制掩码m指定，m指示哪些特征受约束。我们根据控制掩码将带噪运动用目标姿态特征进行插补（即覆盖）为 $\tilde{\mathbf{x}}_t = \mathbf{m} \odot \mathbf{x}_{\text{tgt}} + (1 - \mathbf{m}) \odot \mathbf{x}_t$，其中 $\odot$ 是逐元素乘积[33]。最后，我们将控制掩码与插补后的运动沿特征维度拼接，生成最终输入令牌 $\mathbf{x}_{\text{in}} = [\tilde{\mathbf{x}}_t; \mathbf{m}]$。当未指定约束时，掩码全为零，且 $\tilde{\mathbf{x}}_t = \mathbf{x}_t$。

去噪器预测为 $\hat{\mathbf{x}}_0 = \mathcal{D}_\theta(\mathbf{x}_{\text{in}}, C, t)$，其中 $C = \{\mathbf{c}_{\text{text}}, \mathbf{c}_{\text{dir}}, \mathbf{c}_{\text{extra}}\}$ 是额外条件令牌的集合。$\mathbf{c}_{\text{text}} \in \mathbb{R}^{4096}$ 是输入文本提示 [1] 的LLM2Vec嵌入，我们在早期实验中发现其优于CLIP[29]和T5[30]等常见替代方案。$\mathbf{c}_{\text{dir}} \in \mathbb{R}^2$ 是第一帧的期望朝向方向。最后，$\mathbf{c}_{\text{extra}} \in \mathbb{R}^{P \times 4096}$ 包含 P 个额外的全零令牌。虽然这些额外令牌不能严格视为“寄存器”令牌[4]，因为它们不是学习得到的，但它们在实验中（第6节）达到了增强模型表示能力的类似效果。实践中，我们使用 $P = 49$。所有姿态和条件令牌嵌入到相同维度，并在进入Transformer前加上正弦位置编码。

架构。如图9所示，去噪器分解为两个Transformer编码器：一个用于预测根运动，另一个用于身体运动。根去噪器首先预测全局根运动 $\hat{\mathbf{r}}_0^{\text{glob}}$。尽管只预测根，该去噪器以完整带噪运动 $\mathbf{x}_{\text{in}}$ 为条件，使得输出可以

coordinated with the body motion and any relevant constraints. After the first stage, the global root prediction is transformed into a local representation $\hat{\mathbf{r}}_0^{\text{local}}$, concatenated with the body features from $\mathbf{x}_{\text{in}}$, and inputted to the second transformer to predict the body motion $\hat{\mathbf{b}}_0$. The final output of the denoiser is the concatenated global root and body predictions $\hat{\mathbf{x}}_0 = [\hat{\mathbf{r}}_0^{\text{glob}}; \hat{\mathbf{b}}_0]$. In practice, both transformers use the same architecture with 16 layers, 8 heads, and a latent size of 1024, totaling 282 M learnable parameters in our full model.

While the global root representation is advantageous when modeling root motion, we found that using a *local* root representation is more effective when conditioning the second stage of the model to denoise body motion. Inspired by Guo *et al*. [8], we define the local root pose as $\mathbf{r}^{\text{local}} = [\dot{\mathbf{r}}^a, \dot{\mathbf{r}}^p_{xz}, \mathbf{r}^p_y]$ where $\dot{\mathbf{r}}^a \in \mathbb{R}$ is the angular velocity of the heading, $\dot{\mathbf{r}}^p_{xz} \in \mathbb{R}^2$ is the planar translation velocity of the root, and $\mathbf{r}^p_y \in \mathbb{R}$ is the absolute height. Converting from the global to local root representation is straightforward using finite differences to estimate velocities.

Discussion. The two-stage denoiser design is key to maximizing motion quality and control accuracy together. The global root representation used in the first stage enables conditioning on sparse global root constraints through direct imputation, while the second stage benefits from being conditioned on a more invariant local root representation. Moreover, we hypothesize that body motion prediction is an easier problem when conditioned on the root motion. This is supported experimentally in Sec. 6, where ablations of the two-stage model and the local root representation result in worse performance. Note that unlike prior two-stage approaches, our two-stage model is *interleaved* since the root and body predictions are made at every denoising step. Prior work performs the root denoising process in full before generating the body motion [14, 12], while our model enables root and body corrections throughout denoising to attain better alignment in the end.

The need for the initial heading token $\mathbf{c}_{\text{dir}}$ as input stems from our global representation of joint rotations. To generalize more effectively with this global representation, we speculate that it is helpful for the model to see a wide distribution of rotations during training. Therefore, we choose not to canonicalize motions relative to the heading at the first frame, as is common practice, and instead apply a random first-frame heading augmentation to motions during training (see Sec. 4.3). This means the model can generate motion that starts at any arbitrary heading, which we aim to control at test time with $\mathbf{c}_{\text{dir}}$. The choice not to canonicalize the motion and use $\mathbf{c}_{\text{dir}}$ is also practically convenient at test time. When the model is conditioned on global constraints in the scene or the user wants to generate motion from a sequence of multiple prompts (*e.g.*, from an editing timeline), there is no need to apply a canonicalizing transform before generation and subsequently an inverse transform afterwards, since we can simply specify the desired initial heading with $\mathbf{c}_{\text{dir}}$.

4.3. Training

The denoiser is trained according to the DDPM [11] framework using a modified version of the simplified loss function. At each training iteration, a ground truth motion $\mathbf{x}_0$ is sampled from the dataset and a diffusion timestep $t \sim \mathcal{U}\{1, \ldots, T\}$ and Gaussian noise $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ are used to noise the motion to $\mathbf{x}_t = q(\mathbf{x}_t \mid \mathbf{x}_0, t)$. After obtaining the denoiser prediction $\hat{\mathbf{x}}_0$, the loss is computed on each component of the motion representation:

$$\begin{aligned}\mathcal{L} = \gamma_1||\hat{\mathbf{r}}^p_0 - \mathbf{r}^p_0||_1 + \gamma_2||\hat{\mathbf{r}}^a_0 - \mathbf{r}^a_0||_1 + \gamma_3||\hat{\mathbf{j}}^p_0 - \mathbf{j}^p_0||_1 + \gamma_4||\hat{\mathbf{j}}^v_0 - \mathbf{j}^v_0||_1 \\ + \gamma_5||\hat{\mathbf{j}}^a_0 - \mathbf{j}^a_0||_1 + \gamma_6||\hat{\mathbf{f}}_0 - \mathbf{f}_0||_1 + \gamma_7||\text{FK}(\hat{\mathbf{j}}^a_0) - \mathbf{j}^p_0||_1.\end{aligned} \tag{1}$$

where $||\cdot||_1$ is a smooth L1 loss that uses an L2 term when the loss is small and an L1 term otherwise [7]. FK$(\cdot)$ is the forward kinematics function, which computes joint positions from joint rotations. In practice, the losses are weighted as $\gamma_1 = \gamma_3 = \gamma_5 = 10.0$, $\gamma_2 = 2.0$, $\gamma_4 = 3.0$, $\gamma_6 = 4.0$, and $\gamma_7 = 5.0$. Training uses variable length sequences within each batch, so loss functions are masked accordingly. We use $T = 1000$ diffusion steps for training.

To improve training stability, we use the Adam-atan2 optimizer [5] with a learning rate of $2e{-}5$. Ground truth motions are cropped to a maximum length of 10 sec and translated such that the root position is above the

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与身体运动及任何相关约束相协调。在第一阶段之后，全局根运动预测被转换为局部表示 $\hat{\mathbf{r}}_0^{\text{local}}$ ，与来自 $\mathbf{x}_{\text{in}}$ 的身体特征拼接，并输入第二个变换器以预测身体运动 $\widehat{\mathbf{b}}_0$ 。去噪器的最终输出是拼接后的全局根和身体预测 $\widehat{\mathbf{x}}_0 = \left[\hat{\mathbf{r}}_0^{\text{glob}}; \widehat{\mathbf{b}}_0\right]$ 。实际上，两个变换器使用相同的架构，具有16层、8个头和1024的潜在大小，在我们的完整模型中总计282M可学习参数。

虽然全局根表示在建模根运动时具有优势，但我们发现，在条件化模型第二阶段去噪身体运动时，使用局部根表示更为有效。受Guo等人[8]的启发，我们将局部根姿态定义为 $\mathbf{r}^{\text{local}} = [\dot{\mathbf{r}}^a, \dot{\mathbf{r}}^p_{xz}, \mathbf{r}^p_y]$ ，其中 $\dot{\mathbf{r}}^a \in \mathbb{R}$ 是航向的角速度， $\dot{\mathbf{r}}^p_{xz} \in \mathbb{R}^2$ 是根部的平面平移速度， $\mathbf{r}^p_y \in \mathbb{R}$ 是绝对高度。从全局到局部根表示的转换可以通过有限差分估计速度来直接实现。

讨论。两阶段去噪器设计是同时最大化运动质量和控制精度的关键。第一阶段使用的全局根表示通过直接插补实现对稀疏全局根约束的条件化，而第二阶段则受益于对更不变的局部根表示的条件化。此外，我们假设在条件化于根运动的情况下，身体运动预测是一个更简单的问题。这在第6节中得到了实验支持，其中两阶段模型和局部根表示的消融导致性能下降。注意，与先前的两阶段方法不同，我们的两阶段模型是交错的，因为根和身体预测在每个去噪步骤中进行。先前的工作在生成身体运动之前完整执行根去噪过程[14, 12]，而我们的模型在整个去噪过程中允许根和身体的修正，以实现最终更好的对齐。

需要初始航向标记 $\mathbf{c}_{\text{dir}}$ 作为输入源于我们对关节旋转的全局表示。为了更有效地泛化这种全局表示，我们推测模型在训练期间看到广泛的旋转分布是有帮助的。因此，我们选择不按照常见做法将运动相对于第一帧的航向进行规范化，而是在训练期间对运动应用随机的第一帧航向增强（见第4.3节）。这意味着模型可以生成从任意航向开始的运动，我们旨在测试时通过 $\mathbf{c}_{\text{dir}}$ 控制这一点。不规范化运动并使用 $\mathbf{c}_{\text{dir}}$ 的选择在测试时也实际方便。当模型被场景中的全局约束条件化，或者用户希望从多个提示的序列（例如，从编辑时间线）生成运动时，无需在生成前应用规范化变换，随后再应用逆变换，因为我们可以简单地通过 $\mathbf{c}_{\text{dir}}$ 指定所需的初始航向。

4.3. 训练

去噪器根据DDPM [11]框架，使用简化损失函数的修改版本进行训练。在每次训练迭代中，从数据集中采样一个真实运动 $\mathbf{x}_0$ ，并使用扩散时间步 $t \sim \mathcal{U}\{1, \ldots, T\}$ 和高斯噪声 $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 将运动噪声化为 $\mathbf{x}_t = q(\mathbf{x}_t \mid \mathbf{x}_0, t)$ 。获得去噪器预测 $\widehat{\mathbf{x}}_0$ 后，对运动表示的每个分量计算损失：

$$\begin{aligned}\mathcal{L} = \gamma_1||\hat{\mathbf{r}}^p_0 - \mathbf{r}^p_0||_1 + \gamma_2||\hat{\mathbf{r}}^a_0 - \mathbf{r}^a_0||_1 + \gamma_3||\hat{\mathbf{j}}^p_0 - \mathbf{j}^p_0||_1 + \gamma_4||\hat{\mathbf{j}}^v_0 - \mathbf{j}^v_0||_1 \\ + \gamma_5||\hat{\mathbf{j}}^a_0 - \mathbf{j}^a_0||_1 + \gamma_6||\hat{\mathbf{f}}_0 - \mathbf{f}_0||_1 + \gamma_7||\text{FK}(\hat{\mathbf{j}}^a_0) - \mathbf{j}^p_0||_1.\end{aligned} \tag{1}$$

其中 $||\cdot||_1$ 是平滑L1损失，当损失较小时使用L2项，否则使用L1项[7]。FK$(\cdot)$是前向运动学函数，从关节旋转计算关节位置。实践中，损失权重为
$\gamma_1 = \gamma_3 = \gamma_5 = 10.0, \gamma_2 = 2.0, \gamma_4 = 3.0, \gamma_6 = 4.0$ 和 $\gamma_7 = 5.0$ 。训练使用每个批次内的变长序列，因此损失函数相应地进行掩码处理。我们使用 $T = 1000$ 个扩散步骤进行训练。
为提高训练稳定性，我们使用Adam-atan2优化器[5]，学习率为 $2e-5$ 。真实运动被裁剪至最大长度10秒，并平移使得第一帧的根位置位于

origin at the first frame. The heading direction at the first frame is randomized to ensure the model is robust to variations in global rotations. Our best model configuration is trained with a batch size of 2048 across 16 NVIDIA A100 (SXM4-80GB) GPUs and generates motions at 30 fps.

Training Curriculum. The denoiser is trained in two phases. For the first 500k steps (phase 1), the model is trained purely on the text-to-motion task with no constraints given as input. For the second 500k steps (phase 2), the model is trained on a mix of text and kinematic constraints. During phase 2, kinematic constraints are randomly sampled from a set of pre-defined constraint patterns designed to enable specific functionality at test time. These include: full-body joint positions at sparse keyframes, random subsets of hands and feet positions/rotations at sparse keyframes, 2D root position/heading at sparse keyframes, 2D root position/heading on dense paths, and foot contact configuration at sparse keyframes. Two constraint patterns are mixed together 25% of the time, and 10% of the time no constraints are used (leaving only text input). During phase 2, the maximum number of keyframes sampled for sparse constraints increases linearly from 1 to 20, and sampling is biased towards fewer keyframes to reflect common real-world use cases. Dropout with a rate of 0.1 is used during phase 1, but is removed for phase 2 to avoid dropping out conditioning constraints that are directly overwritten to the noisy motion input. During both phases, the text input is dropped 10% of the time to enable classifier-free guidance at test time [10]. Exponential Moving Average (EMA) is applied every 10 steps throughout training with a decay of 0.995 to maintain an average of the denoiser parameters, which is then used at test time.

4.4. Motion Generation

After training, motions are generated using the DDIM [36] inference process with, by default, 100 denoising steps. We leverage a classifier-free guidance approach that decomposes text and constraint conditioning to allow control over each one individually. In particular, the model output at each denoising step is computed as $\hat{\mathbf{x}}_0 = \mathcal{D}_\varnothing + w_{\text{text}}(\mathcal{D}_{\text{text}} - \mathcal{D}_\varnothing) + w_{\text{constr}}(\mathcal{D}_{\text{constr}} - \mathcal{D}_\varnothing)$ where $\mathcal{D}_\varnothing$ is the model output using no text or constraint conditioning, $\mathcal{D}_{\text{text}}$ uses only text conditioning (no constraints), and $\mathcal{D}_{\text{constr}}$ uses only constraint conditioning (no text). By default, we use $w_{\text{text}} = 2$ and $w_{\text{constr}} = 2$, but a user can adjust each to vary the influence of text and constraint conditioning on the model output. Several prior works make use of gradient-based guidance to further improve constraint following at test-time [31, 14], but we found that since our model is already directly conditioned on the constraints, adding gradient-based guidance gave minimal improvement, substantially increased generation time, and was generally unstable and difficult to tune.

Multi-Prompt Sequencing. While Kimodo is trained to take one text prompt as input, it is practically desirable to generate motions for a sequence of prompts (see Sec. 2.1). We achieve this by generating the motion for each prompt in sequence and adding constraints to maintain plausible transitions. After generating motion for the first prompt, the subsequent prompt is generated with an overlap to the first prompt where several full-body keyframe constraints are added to encourage consistent joint positions and accelerations with the previously generated motion. To ensure a smooth transition, we blend the constrained frames that are shared by the first and second prompt after generation.

Motion Post-Processing. In practice, post-processing can be performed on the model outputs to improve the generated motion. Simple foot locking and IK can clean up any undesirable foot skate using the foot contact classification directly from the model output. It is also helpful to perform a short optimization on the output motion to ensure it *exactly* hits the kinematic constraints, which is challenging for the model to achieve. We use these post-processing approaches in our released demo and codebase, but not for the experiments in Sec. 6 to ensure a fair comparison between methods.

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原点之上。第一帧的朝向方向被随机化，以确保模型对全局旋转的变化具有鲁棒性。我们最佳模型配置使用2048的批次大小，在16块NVIDIA A100（SXM4-80GB）GPU上训练，并以30 fps生成运动。

训练课程。去噪器分两个阶段训练。在前500k步（阶段1），模型仅针对文本到运动任务进行训练，不输入任何约束。在第二个500k步（阶段2），模型在文本和运动学约束的混合数据上训练。在阶段2中，运动学约束从一组预定义的约束模式中随机采样，这些模式旨在测试时实现特定功能。这些包括：稀疏关键帧处的全身关节位置，稀疏关键帧处的手和脚位置/旋转的随机子集，稀疏关键帧处的2D根位置/朝向，密集路径上的2D根位置/朝向，以及稀疏关键帧处的脚接触配置。25%的时间混合两种约束模式，10%的时间不使用约束（仅保留文本输入）。在阶段2中，稀疏约束采样的最大关键帧数从1线性增加到20，并且采样偏向于较少的关键帧以反映常见的实际使用场景。阶段1使用0.1的丢弃率，但阶段2中移除丢弃，以避免丢弃直接覆盖到噪声运动输入上的条件约束。在两个阶段中，文本输入有 10% 的时间被丢弃，以便在测试时实现无分类器引导[10]。在整个训练过程中，每10步应用指数移动平均（EMA），衰减为0.995，以维护去噪器参数的平均值，该平均值在测试时使用。

4.4. 运动生成

训练后，使用DDIM [36]推理过程生成运动，默认采用100步去噪。我们利用无分类器引导方法，将文本和约束条件分解，以便分别控制每一项。具体而言，每个去噪步骤的模型输出计算为 $\hat{\mathbf{x}}_0 = \mathcal{D}_\varnothing + w_{\text{text}} (\mathcal{D}_{\text{text}} - \mathcal{D}_\varnothing) + w_{\text{constr}} (\mathcal{D}_{\text{constr}} - \mathcal{D}_\varnothing)$，其中 $\mathcal{D}_\varnothing$ 为不使用文本或约束条件的模型输出，$\mathcal{D}_{\text{text}}$ 仅使用文本条件（无约束），$\mathcal{D}_{\text{constr}}$ 仅使用约束条件（无文本）。默认情况下，我们使用 $w_{\text{text}} = 2$ 和 $w_{\text{constr}} = 2$，但用户可调整各自以改变文本和约束条件对模型输出的影响。先前多项工作利用基于梯度的引导在测试时进一步改善约束遵循[31, 14]，但我们发现，由于模型已直接基于约束条件进行条件化，添加基于梯度的引导带来的改进微乎其微，却显著增加生成时间，且通常不稳定、难以调参。

多提示序列化。虽然Kimodo训练时接受单个文本提示作为输入，但实际应用中需要为一系列提示生成运动（见第2.1节）。我们通过按顺序为每个提示生成运动，并添加约束以保持合理的过渡来实现。在生成第一个提示的运动后，后续提示与第一个提示有重叠生成，并添加若干全身关键帧约束，以鼓励与先前生成运动的关节位置和加速度保持一致。为确保平滑过渡，我们在生成后将第一个和第二个提示共享的受约束帧进行混合。

运动后处理。实践中，可对模型输出进行后处理以改善生成的运动。简单的脚部锁定和IK可利用模型输出直接提供的脚部接触分类来清理任何不良的脚部滑动。对输出运动进行短暂优化以确保其精确满足运动学约束也很有帮助，这对模型而言具有挑战性。我们在发布的演示和代码库中使用了这些后处理方法，但在第6节的实验中未使用，以确保方法间的公平比较。

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5. Related Work

Our approach is closely related to previous works in generative human motion modeling, while improving upon them in key ways. Besides explicit motion diffusion [42, 52], several alternative approaches have proven successful at text-conditioned human motion generation. Latent motion diffusion [3] denoises motion in a learned latent space rather than the explicit pose space, thereby improving efficiency. In a similar spirit, MMM [26] and MoMask [9] learn a discretized latent space via a VQ-VAE [44], then train a model to generate a sequence of latents through a progressive masked prediction procedure. A different class of methods treat discretized latents as tokens and autoregressively predict them in sequence, similar to large language models [50, 13]. Such latent approaches focus primarily on text control. Those that do handle kinematic constraints require latent test-time optimization to achieve high accuracy [45, 27].

Our choice of explicit motion diffusion is motivated by the ease of controllability on the pose features. OmniControl [47] demonstrated dense and sparse positional control with a ControlNet [51] fine-tuned on top of a base motion diffusion model. GMD [14] uses a combination of imputation and test-time guidance to handle potentially sparse positional constraints. Learning an RL policy on top of a diffusion model has also been explored for kinematic controls [35, 54]. Our method is most related to methods that use imputation to condition generation by directly overwriting constraints in the input motion [34, 12]. CondMDI uses the same imputation approach that we do with a concatenated mask as input to the denoiser [33].

Our method, Kimodo, supports a suite of kinematic controls that is more extensive than prior works. Through direct imputation during training, our denoiser can handle both sparse and dense constraints on positions and joint rotations. This is achieved without additional ControlNet fine-tuning, test-time guidance/optimization, or RL. Moreover, our smoothed root representation lends itself to root constraints commonly found in motion editing applications. While prior work has leveraged a global joint position representation to handle sparse constraints [14, 33, 12, 22], we also adopt the global representation for joint rotations, enabling sparse rotation control. Additionally, previous approaches that use a two-stage model [14, 12] train each stage independently, while our interleaved two-stage denoiser trains end-to-end.

A major challenge in learning effective motion generation models is the relative lack of large-scale datasets, with the most common benchmark HumanML3D [8] containing only 30 hours of motion. Several works try to address this through increasingly larger datasets. Motion-X was an early work to collect a dataset with a majority of motions recovered from online video sources, totaling 144 hours of motion [16]. Datasets have continued to grow through a combination of mocap and videos from several sources [19, 17, 46, 53, 38], with the recent MotionMillion dataset containing 2000 hours of motions [6]. While such diverse data is exciting to study the scaling properties of motion generation, these datasets rely heavily on motions reconstructed from monocular videos and/or mocap from a variety of sources with different framerates and skeletons. Additionally, text descriptions are often labeled through automatic LLM-driven approaches. As a result, the average motion and text quality is considerably lower than the 700 hours of optical mocap data and human-labeled text annotations that Kimodo is trained on. An exciting future direction is how to leverage large-scale video data without compromising the motion quality that can be learned from mocap.

6. Quantitative Evaluation

In this section, we perform a detailed experimental analysis of key design decisions of our approach (Sec. 6.2) and scaling behavior in terms of data size, model size, and batch size (Sec. 6.3).

Most prior works evaluate on the public HumanML3D benchmark [8], which is relatively small scale and is becoming increasingly saturated, as indicated by better-than-ground-truth metrics reported for recent methods. For our experiments, we choose to exclusively use the large-scale, high-quality Bones Rigplay [2] mocap dataset described in Sec. 3, as it provides a robust benchmark for highlighting differences between ablations and variations of our method.

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5. 相关工作

我们的方法与先前在生成式人体运动建模方面的工作密切相关，但在关键方面有所改进。除了显式运动扩散 [42, 52] 之外，几种替代方法已被证明在文本条件的人体运动生成中是成功的。潜在运动扩散 [3] 在学习的潜在空间而非显式姿态空间中对运动进行去噪，从而提高了效率。类似地，MMM [26] 和 MoMask [9] 通过 VQ-VAE [44] 学习离散的潜在空间，然后训练模型通过渐进式掩码预测过程生成潜在序列。另一类方法将离散潜在视为标记，并像大型语言模型 [50, 13] 一样按顺序自回归预测它们。这类潜在方法主要关注文本控制。那些处理运动学约束的方法需要潜在测试时优化才能实现高精度 [45, 27]。

我们选择显式运动扩散是出于对姿态特征可控性的便利考虑。OmniControl [47] 展示了通过 ControlNet [51] 在基础运动扩散模型之上微调来实现密集和稀疏位置控制。GMD [14] 结合了插补和测试时引导来处理可能稀疏的位置约束。在扩散模型之上学习强化学习策略也已被探索用于运动学控制 [35, 54]。我们的方法与通过直接覆盖输入运动中的约束来条件生成的方法最相关 [34, 12]。CondMDI 使用了与我们相同的插补方法，并将拼接的掩码作为去噪器的输入 [33]。

我们的方法 Kimodo 支持比先前工作更广泛的运动学控制套件。通过在训练期间直接插补，我们的去噪器可以处理位置和关节旋转上的稀疏及密集约束。这无需额外的 ControlNet 微调、测试时引导/优化或强化学习即可实现。此外，我们平滑的根表示适用于运动编辑应用中常见的根约束。虽然先前工作利用全局关节位置表示来处理稀疏约束 [14, 33, 12, 22]，我们也采用全局表示来处理关节旋转，从而实现稀疏旋转控制。此外，先前使用两阶段模型的方法 [14, 12] 独立训练每个阶段，而我们的交错两阶段去噪器是端到端训练的。

学习有效运动生成模型的一个主要挑战是相对缺乏大规模数据集，最常见的基准 HumanML3D [8] 仅包含 30 小时的运动数据。几项工作试图通过越来越大的数据集来解决这个问题。Motion-X 是早期收集数据集的工作，其中大部分运动来自在线视频源，总计 144 小时的运动数据 [16]。数据集通过结合动作捕捉和多个来源的视频持续增长 [19, 17, 46, 53, 38]，最近的 MotionMillion 数据集包含 2000 小时的运动数据 [6]。虽然这些多样化的数据对于研究运动生成的扩展特性令人兴奋，但这些数据集严重依赖于从单目视频和/或来自不同帧率和骨架的多种来源的动作捕捉重建的运动。此外，文本描述通常通过自动的 LLM 驱动方法进行标注。因此，平均运动和文本质量远低于 Kimodo 训练所用的 700 小时光学动作捕捉数据和人工标注文本。一个令人兴奋的未来方向是如何利用大规模视频数据而不损害可从动作捕捉中学到的运动质量。

6. 定量评估

在本节中，我们对方法的关键设计决策（第6.2节）以及数据规模、模型规模和批处理规模方面的扩展行为（第6.3节）进行了详细的实验分析。

大多数先前的工作在公开的HumanML3D基准[8]上进行评估，该基准规模相对较小，且正变得越来越饱和，这从近期方法报告的超真实度指标中可以看出。在我们的实验中，我们选择专门使用第3节中描述的大规模、高质量的Bones Rigplay [2]动作捕捉数据集，因为它为突出我们方法消融和变体之间的差异提供了稳健的基准。

6.1. Experiment Setting

Dataset. We hold out 10% of the motions in Rigplay from training for evaluation. Splits are determined based on unique behaviors (*i.e*., action types) in the dataset, such that the test set contains only novel behaviors that were not seen during training. For results reported here, we evaluate on a subset containing about 5k motions that cover all distinct behaviors in the test set. For these experiments, models are trained on the native 27-joint skeleton version of the dataset.

Test Cases. Models are evaluated in three different settings. (1) To evaluate the text-to-motion task with no kinematic constraints, we prompt the model with the high-level *Overview Prompt* for each motion (see Sec. 3). (2) We also evaluate unconstrained text-to-motion with a *Fine-Grained Prompt* that tends to describe a shorter motion containing an atomic action. (3) To evaluate combined text+constraint conditioning, we aggregate results from a test suite containing diverse variations of kinematic constraints sampled from the ground truth motion. Constraints in this suite reflect real-world use cases including sparse full-body joint position keyframes, in-betweening with blocks of keyframes at the start and/or end of a motion, sparse keyframes on position/rotation of end-effectors (hands and feet), sparse 2D waypoint keyframes for the root, dense 2D root paths, and mixing multiple constraints together. Constrained generation is evaluated at motion lengths from 3 to 9 sec, both with and without text-conditioning. For a subset of test cases, we apply random perturbations on the global translation and heading of constraints to evaluate generalization.

Evaluation Metrics. We adapt common metrics from prior work [8]. To evaluate text-following, we report *Top-3 R-precision (R@3)* using a TMR embedding model [25] trained on the full Rigplay dataset, including both train and test split. Notably, retrieval with TMR is performed over the entire test set rather than within small batches of 32 as is common in prior work, significantly increasing the challenge. For motion quality, we measure *FID* using the same TMR model. We also report a *Foot Skate* metric that computes the mean velocity of feet joints in the output motion for frames where the model predicts the joint should be in static contact with the ground. This foot skate metric depends on the *Foot Contact Classification Accuracy*, which is generally very good across all methods, but is also reported for completeness.

For constraint-conditioned generation, the *average distance error* between the input constraint and the generated motion at constrained frames is reported. Errors are split into full-body positions, end-effector position/rotation, and 2D root position and averaged over all test cases. Because our model uses a smoothed root representation, the model can generate motion where the smoothed root matches the constraint, but the projected pelvis of the character still deviates from the smooth path/waypoint constraint. As discussed in Sec. 4.1, this flexibility is useful for maintaining natural motion when following root constraints, however, significant deviation of the pelvis from the constraint is undesirable as the character can appear to drift from the constraint path. To evaluate the extent of this deviation, we report the *95th percentile of the 2D position error* between the smoothed root constraint and projected pelvis. This value indicates the maximum that the pelvis will stray from smoothed root constraints for the vast majority of generated motions; closest to ground truth is optimal.

Model Setting. Unless otherwise noted, the models presented in these experiments are trained at 20 fps using 8 GPUs (resulting in a batch size of 1024 motions) and on data with all the augmentations described in Sec. 3. This differs slightly from our best models demonstrated in Sec. 2, which are 30 fps and trained with 16 GPUs, but the trends are still informative.

6.2. Ablation Study

In Tab. 1, we compare the full model architecture and training curriculum to several strong baselines to justify key design decisions. Metrics computed on the ground truth data are shown for reference.

Two-Stage Denoiser. We first compare our decomposed two-stage denoiser design to a *One-Stage* baseline that uses a single transformer to simultaneously denoise root and body motion. To ensure a fair comparison,

6.1. 实验设置

数据集。我们从Rigplay中留出10%的动作用于评估，不参与训练。划分基于数据集中独特的行为（即动作类型），确保测试集仅包含训练中未见的新行为。对于此处报告的结果，我们在包含约 5k 个动作的子集上进行评估，这些动作覆盖了测试集中的所有不同行为。在这些实验中，模型在数据集的原生27关节骨架版本上训练。

测试用例。模型在三种不同设置下进行评估。(1) 为评估无运动学约束的文本到动作任务，我们使用每个动作的高层概述提示（见第3节）来提示模型。(2) 我们还使用细粒度提示评估无约束的文本到动作，该提示倾向于描述包含原子动作的较短动作。(3) 为评估文本+约束联合条件，我们汇总了从真实动作中采样的多种运动学约束变体的测试套件结果。该套件中的约束反映了实际使用场景，包括稀疏全身关节位置关键帧、在动作开始和/或结束处带有关键帧块的中间帧插值、末端执行器（手和脚）位置/旋转的稀疏关键帧、根节点的稀疏2D路径点关键帧、密集2D根路径，以及多种约束的混合。约束生成在3至9秒的动作长度下评估，同时考虑有无文本条件。对于部分测试用例，我们对约束的全局平移和朝向施加随机扰动以评估泛化能力。

评估指标。我们改编了先前工作的常用指标[8]。为评估文本跟随能力，我们使用在完整Rigplay数据集（包括训练和测试划分）上训练的TMR嵌入模型[25]报告Top-3 R精度（R@3）。值得注意的是，TMR检索在整个测试集上进行，而非像先前工作那样在32的小批次内进行，这显著增加了挑战性。对于动作质量，我们使用相同的TMR模型测量FID。我们还报告了脚滑指标，该指标计算输出动作中模型预测关节应与地面静态接触的帧上脚关节的平均速度。此脚滑指标依赖于脚接触分类准确率，该准确率在所有方法中通常都非常好，但为完整性也一并报告。

对于约束条件生成，报告输入约束与生成动作在约束帧上的平均距离误差。误差分为全身位置、末端执行器位置/旋转和2D根位置，并在所有测试用例上取平均。由于我们的模型使用平滑根表示，模型可以生成平滑根与约束匹配的动作，但角色的投影骨盆仍可能偏离平滑路径/路径点约束。如第4.1节所述，这种灵活性在遵循根约束时有助于保持自然动作，然而骨盆显著偏离约束是不可取的，因为角色可能看起来偏离约束路径。为评估这种偏离程度，我们报告平滑根约束与投影骨盆之间2D位置误差的第95百分位数。该值表示在绝大多数生成动作中骨盆偏离平滑根约束的最大程度；最接近真实值者为最优。

模型设置。除非另有说明，这些实验中呈现的模型在20 fps下使用8个GPU训练（产生1024个动作的批次大小），并使用第3节所述的所有增强数据。这与第2节中展示的最佳模型略有不同，后者为30 fps并使用16个GPU训练，但趋势仍具有参考价值。

6.2. 消融研究

在表1中，我们将完整模型架构和训练课程与几个强基线进行比较，以证明关键设计决策的合理性。参考性地展示了基于真实数据计算的指标。

两阶段去噪器。我们首先将分解的两阶段去噪器设计与使用单一变压器同时去噪根部和身体动作的单阶段基线进行比较。为确保公平比较，

Method	Text-Following Evaluation: Overview Prompt Test Set R@3↑	FID ↓	Skate (cm/s)↓	Contact↑	Fine-Grained Prompt Test Set R@3↑	FID↓	Skate (cm/s)↓	Contact↑	Constrained Evaluation: Full-Body Pos (cm)↓	End-Effector Joints Pos (cm)↓	End-Effector Joints Rot (deg)↓	2D Root Pos (cm)↓	2D Pelvis Pos@95% (cm)
Ground Truth	75.6	0.0	2.21	1.00	79.4	0.00	2.23	1.00	-	-	-	-	6.3
Full Model (Ours)	**71.9**	1.85	**3.87**	**0.98**	63.5	1.67	**3.88**	**0.98**	2.67	3.09	4.18	2.90	9.7
One-Stage Arch	71.5	**1.65**	7.59	0.94	63.5	**1.51**	6.80	0.95	8.37	10.19	5.19	7.74	21.1
Second Stage Global	70.3	1.87	4.17	**0.98**	63.2	1.66	4.07	**0.98**	2.97	3.39	5.67	3.25	10.2
No Smoothed Root	71.6	1.75	4.39	0.97	**64.0**	1.55	4.27	**0.98**	2.68	3.19	**3.93**	3.21	7.9
No Extra Tokens	70.9	1.95	4.28	0.97	61.6	1.77	4.17	**0.98**	**2.40**	**2.59**	5.55	**2.85**	9.18
No Train Curriculum	71.3	1.84	3.92	**0.98**	63.2	1.66	3.91	**0.98**	5.80	6.59	4.34	5.71	15.5

Table 1: **Ablation Study.** Evaluation of text and constraint-conditioned motion generation on the Rigplay test set. The full model is compared to various baselines to justify key design decisions, including the two-stage denoiser, smoothed root representation, and dual-phase training curriculum. All models are trained using a medium batch size (8 GPU) at 20 fps. FID is multiplied $\times 100$ for readability.

we increase the number of layers and latent size of the baseline such that the number of learnable parameters is similar to the two-stage model. While the text-following capabilities of this baseline are about the same as the full model, we note a substantial increase in foot skating, indicating that generating body motion conditioned on root motion is indeed easier than generating both simultaneously. The one-stage baseline also causes a substantial increase in constraint errors.

The *Second Stage Global* baseline uses the global root representation in the second (body) stage of the denoiser instead of converting the input root motion to a local representation. This tends to cause an increase in foot skating, likely due to the lack of invariance of the global root representation.

Smoothed Root Representation. The next baseline directly uses the pelvis joint projected to the ground as the root instead of the smoothed representation used in the full model, which causes notably increased foot skate. Without the smoothed root, body joint positions are represented with respect to the pelvis, which may be more difficult to learn due to the high-frequency motions of the pelvis. Despite this foot skate, constraint accuracy in on par with the smoothed root representation. As expected, the 95th percentile of pelvis error is lower than the full model, since the baseline is directly trained to match the pelvis to the root constraint. However, in practice this can cause qualitatively unnatural motions when constraining motions with straight lines or smoothed curves as is common in animation applications. For example, asking the baseline to generate a walking motion along a straight line results in a motion with a stealthy/sneaking style or old person style, since these minimize lateral pelvis movements and therefore deviate less from the straight line constraint.

Extra “Register" Tokens. The *No Extra Tokens* baseline removes the extra register tokens, leaving only the text embedding and heading token as additional conditioning for the model. This tends to reduce performance particularly for text-following and motion quality, indicating that the added tokens increase the representational capacity of the model.

Dual-Phase Training Curriculum. The *No Train Curriculum* baseline directly trains the model for text+constraint conditioning from scratch, rather than training in two phases, as described in Sec. 4.3. For a fair comparison, the baseline is trained for 1 million steps, the same as the total steps of phased training. During training, the baseline receives text-only or text+constraint inputs with equal probability and does not use dropout. This baseline reaches comparable text-following accuracy and motion quality as using phased training, but sees an increase in constraint errors. In two-phase training, phase 1 is dedicated to pre-training on text-to-motion so that phase 2 can be dedicated to constraint-following, but non-phased training must balance text and constraint learning for the entire duration of training. While it is possible there is a perfect balance between text-only and text+constraint input sampling that will result in competitive performance across the board, we find the phased training works well without additional hyperparameter tuning.

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方法	文本跟随评估：概述提示测试集 R@3 ↑	FID ↓	滑冰 (cm/s) ↓	接触 ↑	细粒度提示测试集 R@3 ↑	FID ↓	滑步 (cm/s) ↓	接触 ↑	约束评估：Full-Body 位置 (cm) ↓	末端执行器关节 位置 (cm) ↓	旋转 (度) ↓	2D 根 位置 (cm) ↓	2D 骨盆 位置@95% (cm)
真实数据	75.6	0.0	2.21	1.00	79.4	0.00	2.23	1.00	-	-	-	-	6.3
完整模型（我们的）	71.9	1.85	3.87	0.98	63.5	1.67	3.88	0.98	2.67	3.09	4.18	2.90	9.7
One-Stage Arch	71.5	1.65	7.59	0.94	63.5	1.51	6.80	0.95	8.37	10.19	5.19	7.74	21.1
第二阶段全局	70.3	1.87	4.17	0.98	63.2	1.66	4.07	0.98	2.97	3.39	5.67	3.25	10.2
无平滑根节点	71.6	1.75	4.39	0.97	64.0	1.55	4.27	0.98	2.68	3.19	3.93	3.21	7.9
无额外标记	70.9	1.95	4.28	0.97	61.6	1.77	4.17	0.98	2.40	2.59	5.55	2.85	9.18
无训练课程	71.3	1.84	3.92	0.98	63.2	1.66	3.91	0.98	5.80	6.59	4.34	5.71	15.5

表1：消融研究。在Rigplay测试集上对文本和约束条件驱动的运动生成进行评估。将完整模型与各种基线进行比较，以验证关键设计决策，包括两阶段去噪器、平滑根节点表示和双阶段训练课程。所有模型均使用中等批量大小（8 GPU）在 20fps 下训练。为便于阅读，FID乘以 $\times 100$ 。

我们增加了基线的层数和潜在尺寸，使其可学习参数数量与两阶段模型相近。虽然该基线的文本跟随能力与完整模型大致相同，但我们注意到脚部滑步显著增加，这表明基于根节点运动生成身体运动确实比同时生成两者更容易。单阶段基线还导致约束误差大幅增加。

第二阶段全局基线在去噪器的第二阶段（身体阶段）使用全局根表示，而不是将输入的根运动转换为局部表示。这往往会导致脚滑步增加，可能是由于全局根表示缺乏不变性。

平滑根表示。下一个基线直接使用投影到地面的骨盆关节作为根，而不是完整模型中使用的平滑表示，这导致脚滑步显著增加。没有平滑的根，身体关节位置相对于骨盆表示，由于骨盆的高频运动，这可能更难学习。尽管存在脚滑步，约束精度与平滑根表示相当。如预期，骨盆误差的第95百分位低于完整模型，因为基线直接训练以匹配骨盆到根约束。然而，在实践中，当约束运动为直线或平滑曲线（这在动画应用中很常见）时，这可能导致定性上不自然的运动。例如，要求基线生成沿直线行走的运动，会导致具有潜行/偷偷摸摸风格或老人风格的运动，因为这些最小化横向骨盆运动，因此偏离直线约束较少。

额外“注册”令牌。无额外令牌基线移除了额外的注册令牌，仅保留文本嵌入和朝向令牌作为模型的额外条件。这往往会降低性能，特别是在文本跟随和运动质量方面，表明添加的令牌增加了模型的表示能力。

双阶段训练课程。无训练课程基线直接从零开始训练模型进行文本+约束条件，而不是如第4.3节所述分两阶段训练。为了公平比较，基线训练了100万步，与分阶段训练的总步数相同。在训练期间，基线以相等概率接收仅文本或文本+约束输入，并且不使用丢弃。该基线达到了与分阶段训练相当的文本跟随准确性和运动质量，但约束误差增加。在两阶段训练中，第一阶段专门用于文本到运动的预训练，以便第二阶段可以专门用于约束跟随，但非分阶段训练必须在整个训练期间平衡文本和约束学习。虽然可能存在文本仅和文本+约束输入采样的完美平衡，从而在各方面产生竞争性性能，但我们发现分阶段训练无需额外超参数调整即可良好工作。

	Text-Following Evaluation								Constrained Evaluation				
	Overview Prompt Test Set				Fine-Grained Prompt Test Set				Full-Body	End-Effector Joints		2D Root	2D Pelvis
Method	**R@3**↑	**FID** ↓	**Skate** (cm/s)↓	**Contact**↑	**R@3**↑	**FID**↓	**Skate** (cm/s)↓	**Contact**↑	**Pos** (cm)↓	**Pos** (cm)↓	**Rot** (deg)↓	**Pos** (cm)↓	**Pos@95%** (cm)
Ground Truth	75.6	0.00	2.21	1.00	79.4	0.00	2.23	1.00	-	-	-	-	6.3
Full Dataset	**71.5**	1.84	**4.23**	**0.97**	63.0	1.07	**4.15**	**0.98**	**2.77**	**3.31**	**5.36**	**3.29**	10.7
50% Dataset	70.8	**1.81**	4.43	**0.97**	**63.4**	**1.06**	4.27	0.97	3.13	3.56	6.29	3.32	10.7
10% Dataset	71.0	2.07	5.28	0.96	62.4	1.41	5.12	0.97	4.60	6.91	10.03	4.83	15.5
L Model (282 M)	**71.9**	**1.85**	**3.87**	**0.98**	**63.5**	**1.67**	**3.88**	**0.98**	**2.67**	**3.09**	**4.18**	**2.90**	9.7
M Model (148 M)	69.2	2.36	4.45	0.97	60.1	2.12	4.31	0.97	3.26	3.72	4.70	3.34	10.0
S Model (56 M)	64.0	3.10	4.53	0.97	55.5	2.48	4.47	0.97	3.56	3.98	11.27	3.49	10.8
L Batches (16 GPU)	**73.6**	**1.61**	3.97	**0.98**	**63.8**	**1.52**	**3.87**	**0.98**	**2.33**	**2.71**	**4.09**	**2.35**	8.5
M Batches (8 GPU)	71.9	1.85	**3.87**	**0.98**	63.5	1.67	3.88	**0.98**	2.67	3.09	4.18	2.90	9.7
S Batches (4 GPU)	69.4	2.01	4.45	0.97	60.3	1.98	4.27	**0.98**	2.97	3.68	5.61	3.42	10.1

Table 2: **Scaling Analysis.** Evaluation of text and constraint-conditioned motion generation on the Rigplay test set. (Top) Increasing the amount of training data improves motion quality and constraint accuracy due to increased diversity. (Middle) Increased model size improves performance on all metrics. (Bottom) Increasing batch size by using more GPUs generally improves performance across the board.

6.3. Scaling Analysis

Tab. 2 evaluates how scaling affects model performance across three different axes.

Data Size. The top part of the table compares using the full training dataset to training on a subset of 50% and 10% of training motions. Subsets are strategically sampled to include all unique behaviors (action types) from the full dataset, but with the number of performances for each behavior reduced to the desired fraction. As a result, this experiment primarily evaluates how much repeated performances of the same behavior across many actors influences motion generation ability. For this experiment only, we do not use the augmented stitched motions described in Sec. 3.

Comparing the three dataset sizes, we see that foot skate and constraint accuracy monotonically improve with more available training data, with significantly decreased performance when using 10% of the data (*i.e.*, the same order of magnitude as popular AMASS [21] and HumanML3D datasets). Curiously, R-precision and FID are not significantly affected by dataset size, which we believe is an artifact of how we subsample the training data. Since the baselines are trained on the same unique behaviors as the full training set, they should still generate reasonable results for test prompts. Therefore, retrieval with TMR is still successful and R-precision is not greatly affected. Similarly, the generated motion distributions for the test set will be similar even if using only 10% of the data, especially after embedding with TMR and being fit with a unimodal Gaussian, as is done in the FID metric. What R-precision and FID do not capture are the fine-grained differences in motion distribution and subtle but important variations in motion, which manifests as reduced constraint following accuracy due to training on less diverse data.

Model Size. The middle part of Tab. 2 compares large (L), medium (M), and small (S) variants of our model in terms of learnable parameters. Our full best model uses the large size. The medium variant uses 8 layers in the transformer encoders instead of 16, while the small variant additionally decreases the latent size to 512 from 1024. We see that increasing model size improves performance across all metrics. While continuing to increase model size to the order of 500M or 1B parameters could potentially further improve performance, we found training stability becomes a bigger challenge and we speculate that without more data there will be diminishing returns.

Batch Size. The bottom part of the table evaluates how increasing batch size affects performance. In practice, a larger batch size means using more GPUs so we compare small (S), medium (M), and large (L) batch sizes using 4, 8, and 16 GPUs, respectively. This corresponds to 512, 1024, and 2048 batch sizes. As shown in the table, using more GPUs generally improves performance across the board as the gradient estimate during optimization becomes more accurate. We use the large batch size with 16 GPUs to train our best model.

	文本跟随评估								约束评估				
	概述提示测试集				细粒度提示测试集				Full-Body	末端执行器关节		2D 根	2D 骨盆
方法	R@3 ↑	FID ↓	滑步（厘米/秒）↓	接触↑	R@3 ↑	FID ↓	滑行 (cm/s) ↓	接触↑	位置（厘米）↓	位置（厘米）↓	旋转（度）↓	位置（厘米）↓	位置@95% （厘米）
真实数据	75.6	0.00	2.21	1.00	79.4	0.00	2.23	1.00	-	-	-	-	6.3
完整数据集	71.5	1.84	4.23	0.97	63.0	1.07	4.15	0.98	2.77	3.31	5.36	3.29	10.7
50% 数据集	70.8	1.81	4.43	0.97	63.4	1.06	4.27	0.97	3.13	3.56	6.29	3.32	10.7
10% 数据集	71.0	2.07	5.28	0.96	62.4	1.41	5.12	0.97	4.60	6.91	10.03	4.83	15.5
L模型（282M ）	71.9	1.85	3.87	0.98	63.5	1.67	3.88	0.98	2.67	3.09	4.18	2.90	9.7
M模型（148M ）	69.2	2.36	4.45	0.97	60.1	2.12	4.31	0.97	3.26	3.72	4.70	3.34	10.0
S模型（56M ）	64.0	3.10	4.53	0.97	55.5	2.48	4.47	0.97	3.56	3.98	11.27	3.49	10.8
L Batches (16 GPU)	73.6	1.61	3.97	0.98	63.8	1.52	3.87	0.98	2.33	2.71	4.09	2.35	8.5
M Batches (8 GPU)	71.9	1.85	3.87	0.98	63.5	1.67	3.88	0.98	2.67	3.09	4.18	2.90	9.7
S Batches (4 GPU)	69.4	2.01	4.45	0.97	60.3	1.98	4.27	0.98	2.97	3.68	5.61	3.42	10.1

表2：扩展性分析。在Rigplay测试集上对文本和约束条件驱动的运动生成进行评估。（顶部）增加训练数据量通过提高多样性改善了运动质量和约束准确性。（中部）增大模型规模在所有指标上均提升了性能。（底部）通过使用更多GPU增大批处理大小通常能全面改善性能。

6.3. 规模分析

表2评估了规模如何影响模型在三个不同维度上的性能。

数据规模。表格顶部比较了使用完整训练数据集与在 50% 和10%训练动作子集上训练的情况。子集经过策略性采样，以包含完整数据集中的所有独特行为（动作类型），但每种行为的表演次数减少到所需比例。因此，该实验主要评估同一行为在多个演员间的重复表演对动作生成能力的影响程度。仅在此实验中，我们不使用第3节中描述的增强拼接动作。

比较三种数据规模，我们发现随着可用训练数据的增加，脚滑和约束准确性单调提升，使用10%数据时性能显著下降（即与流行的AMASS [21]和HumanML3D数据集数量级相同）。有趣的是，R-精度和FID并未受数据规模显著影响，我们认为这是训练数据子采样方式造成的假象。由于基线模型与完整训练集训练了相同的独特行为，它们仍应为测试提示生成合理结果。因此，使用TMR进行检索仍然成功，R-精度未受太大影响。同样，即使仅使用10%数据，测试集的生成动作分布也会相似，尤其是在TMR嵌入并拟合单峰高斯后（如FID指标所做）。R-精度和FID未能捕捉的是动作分布中的细微差异和动作中微妙但重要的变化，这表现为由于训练数据多样性不足而导致的约束遵循准确性降低。

模型规模。表2中间部分比较了我们模型的大（L）、中（M）、小（S）变体在可学习参数方面的差异。我们完整的最佳模型使用大规模。中等变体在Transformer编码器中使用8层而非16层，而小变体进一步将潜在大小从1024降至512。我们看到增加模型规模在所有指标上均提升性能。虽然继续将模型规模增加到 500M 或10亿参数量级可能进一步提升性能，但我们发现训练稳定性成为更大挑战，并推测没有更多数据将出现收益递减。

批大小。表格底部评估了增加批大小对性能的影响。实践中，更大的批大小意味着使用更多GPU，因此我们分别使用4、8和16个GPU比较小（S）、中（M）和大（L）批大小。这对应512、1024和2048的批大小。如表所示，使用更多GPU通常全面改善性能，因为优化过程中的梯度估计变得更准确。我们使用16个GPU的大批大小来训练最佳模型。

7. Conclusion

We have introduced Kimodo, a kinematic motion diffusion model trained on a large-scale motion capture dataset that generates high-quality human motions and can be controlled through text and a variety of kinematic constraints. Our model enables easy authoring of motions using intuitive interfaces as shown in Sec. 2, and can be applied directly to generating robot motion after retargeting the training dataset. After training entirely on optical mocap data, our model gives a strong foundation for further scaling up of human motion generation.

Future Challenges. Looking forward, a promising direction is to further scale up the model with motions reconstructed from internet videos or generated videos. An important challenge here will be how to combine clean and noisy data sources without compromising output motion quality from the model. While Kimodo is designed specifically for "offline" motion authoring and can take several seconds to generate a motion, applications such as robotics and digital twin simulation require a runtime model that dynamically controls humanoids and reacts to changing environments. To this end, an interesting avenue is moving diffusion to a learned latent space and reformulating motion generation to be an autoregressive problem. Finally, scene and object interactions are crucial to making motion generation models truly practical for most applications. Gathering data for this problem becomes even more challenging, and will require creative solutions.

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7. 结论

我们介绍了Kimodo，一种在大规模动作捕捉数据集上训练的动力学运动扩散模型，能够生成高质量的人体运动，并可通过文本和各种动力学约束进行控制。我们的模型支持使用直观界面轻松创作动作，如第2节所示，并可在重定向训练数据集后直接应用于生成机器人运动。在完全使用光学动作捕捉数据训练后，我们的模型为进一步扩展人体运动生成提供了坚实基础。

未来挑战。展望未来，一个有前景的方向是利用从互联网视频或生成视频中重建的运动进一步扩展模型。这里的一个重要挑战是如何结合干净和嘈杂的数据源，而不损害模型的输出运动质量。虽然Kimodo专为“离线”动作创作设计，生成一个动作可能需要几秒钟，但机器人技术和数字孪生模拟等应用需要能够动态控制人形机器人并响应环境变化的运行时模型。为此，一个有趣的途径是将扩散模型迁移到学习到的潜在空间，并将运动生成重新表述为自回归问题。最后，场景和物体交互对于使运动生成模型在大多数应用中真正实用至关重要。为这一问题收集数据将更具挑战性，需要创造性的解决方案。

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